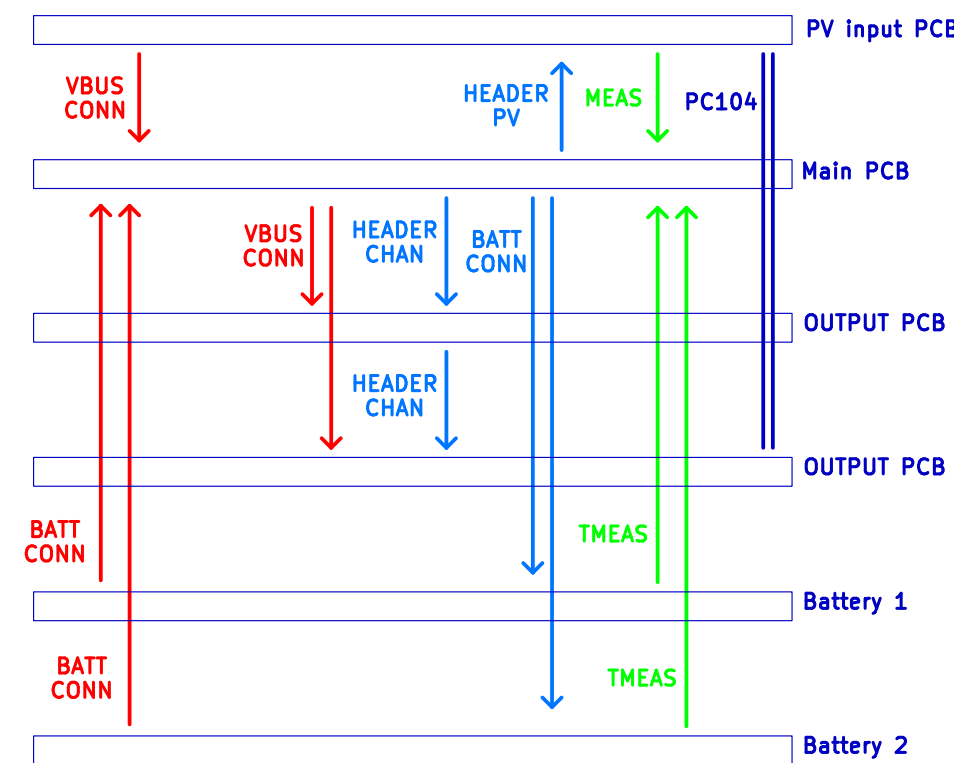
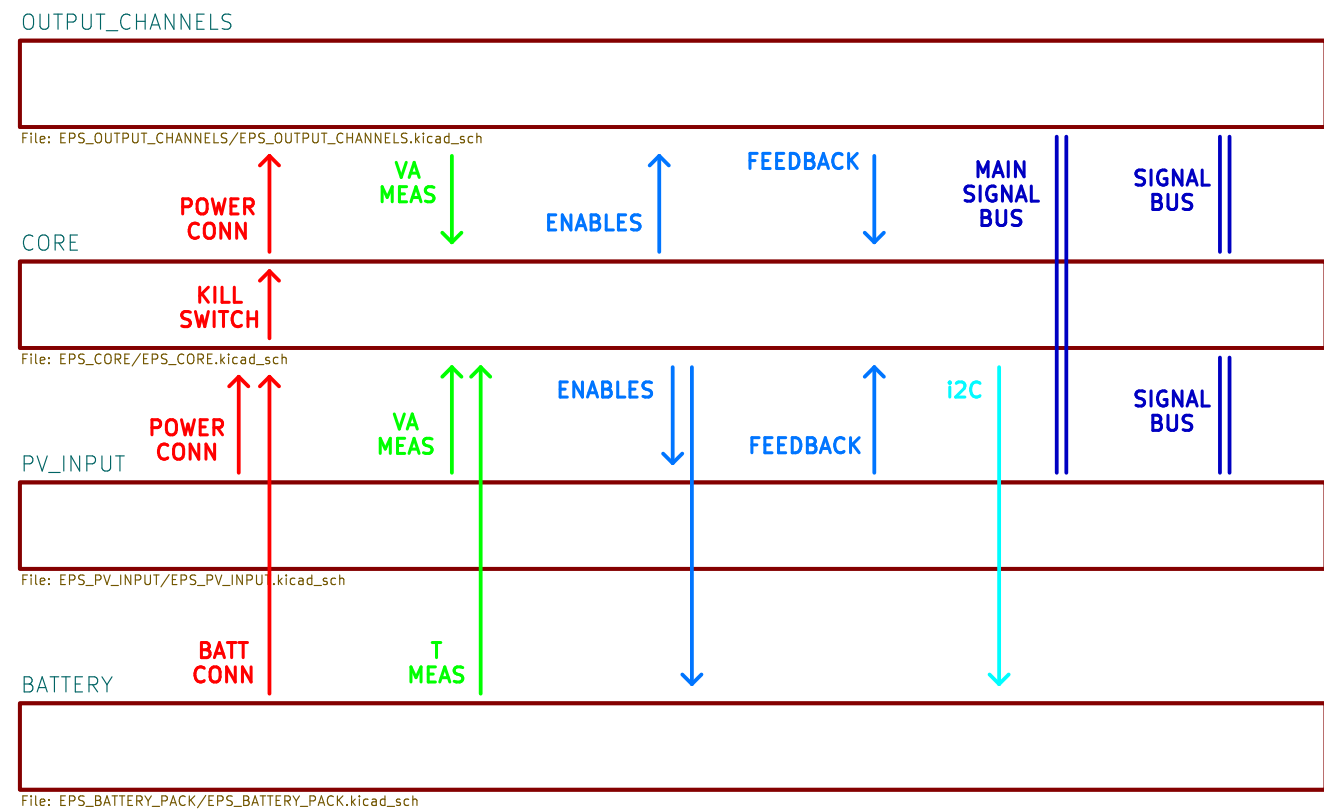
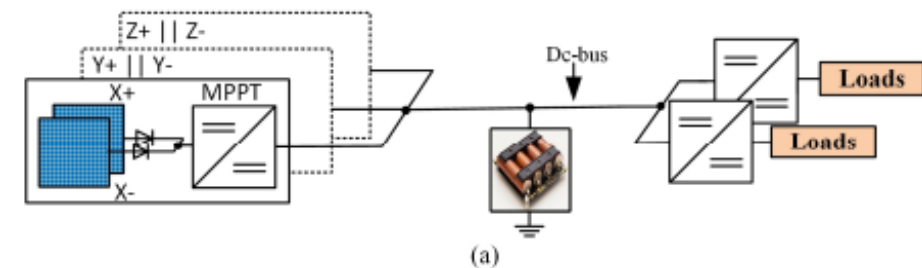
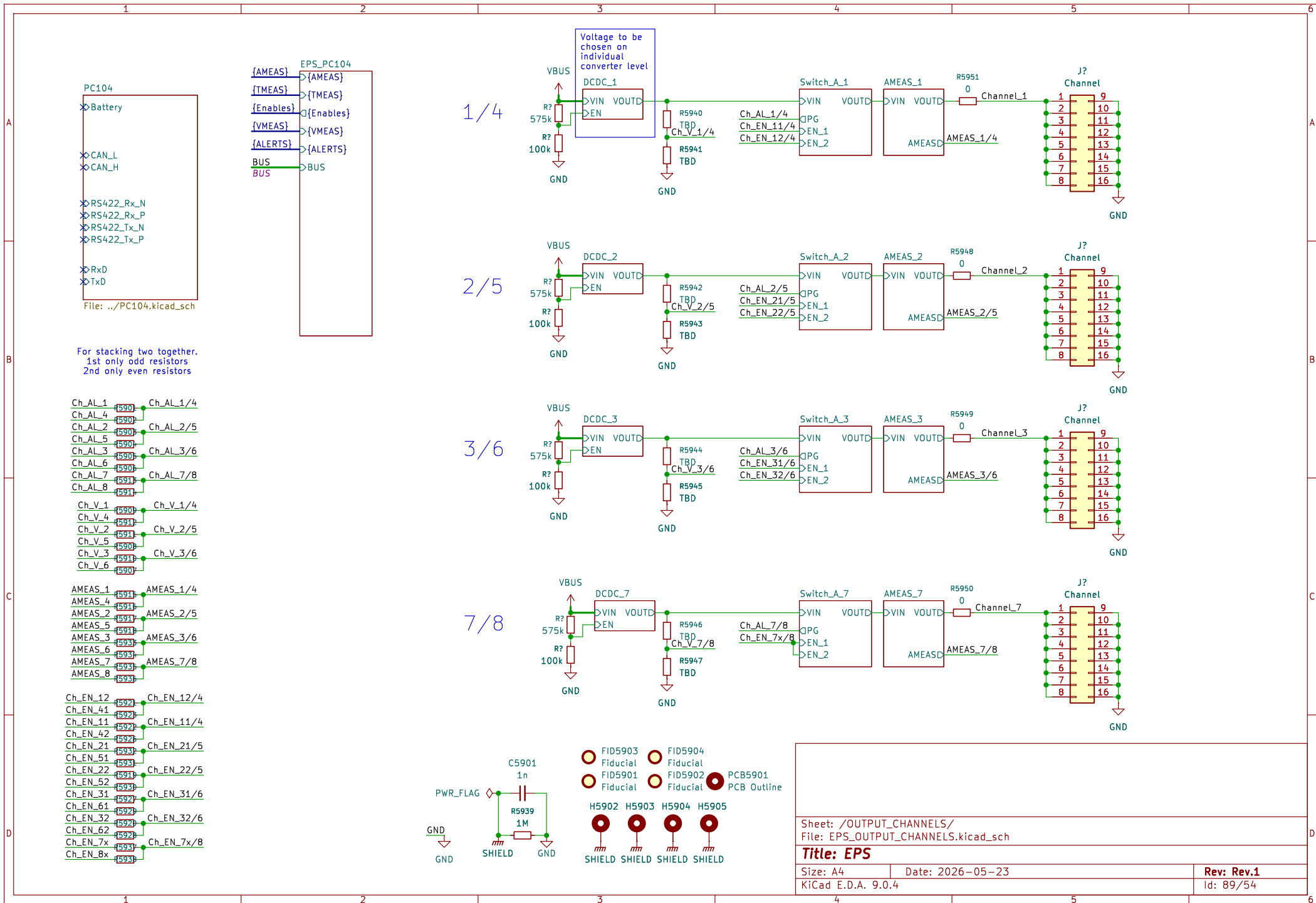


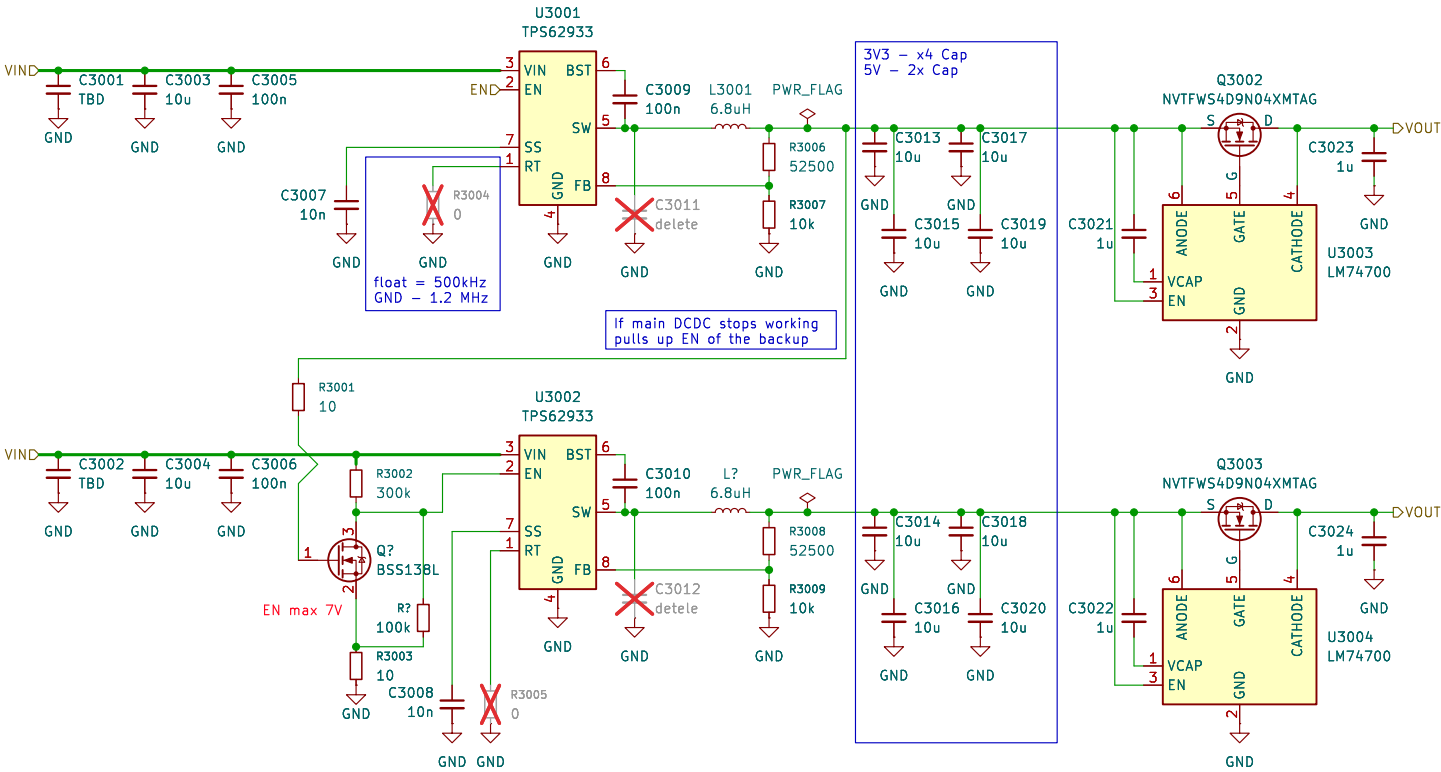


Sheet: /		
File: EPS.kicad_sch		
Title: EPS		
Size: A3	Date: 2026-05-23	Rev: Rev.1
KiCad E.D.A. 9.0.4	Id: 1/54	



Inputs	Outputs
BUS Voltage	Set voltage

Output voltage
5V
 $10000 \cdot (5V - 0.8) / 0.8 \Rightarrow R_{x03}, R_{x00} = 52500$



Sheet: /OUTPUT_CHANNELS/DCDC_1/
File: DCDC_ADJUCTABLE.kicad_sch

Title: EPS

Size: A4

Date: 2026-05-23

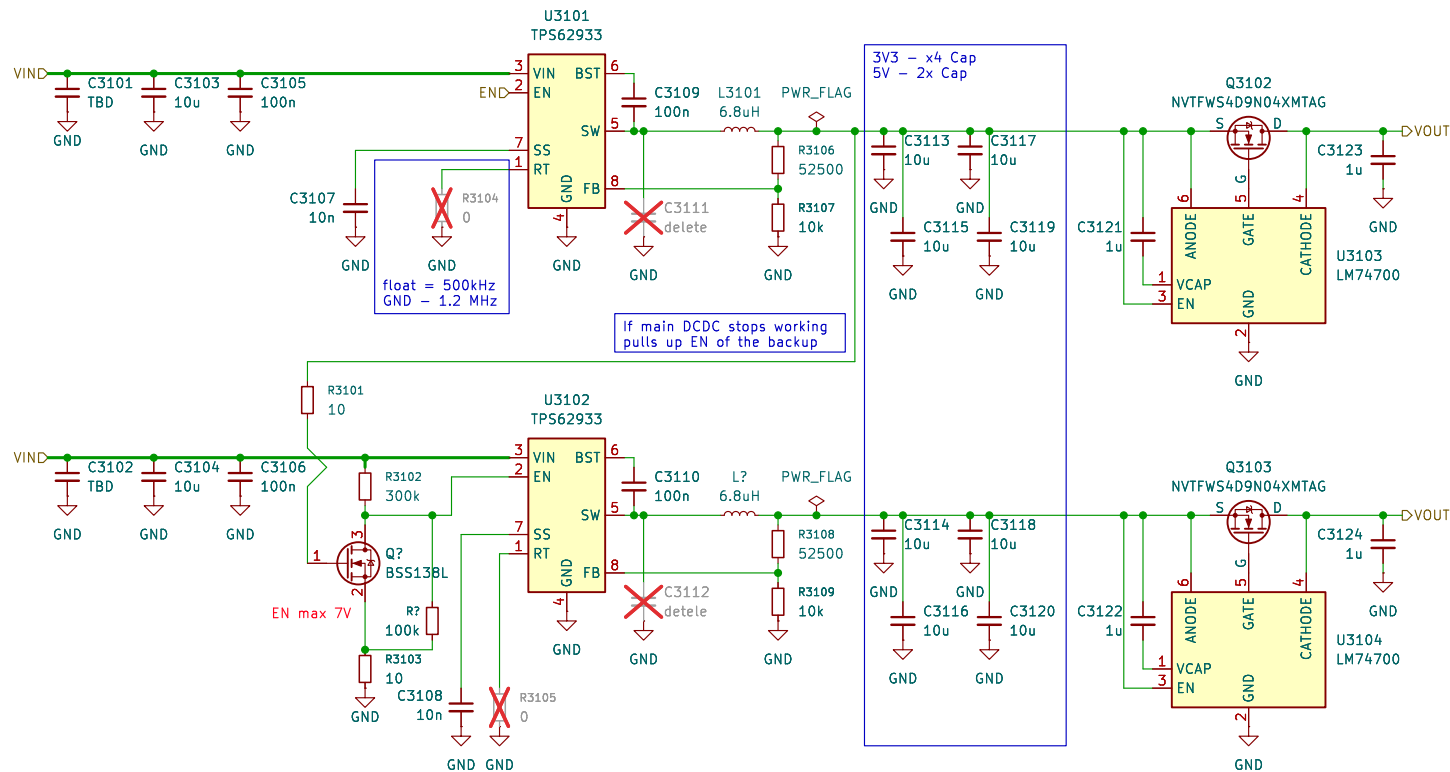
Rev: Rev.1

KiCad E.D.A. 9.0.4

Id: 38/54

Inputs	Outputs
BUS Voltage	Set voltage

Output voltage
5V
 $10000 \cdot (5V - 0,8) / 0,8 \Rightarrow R_{xx03}, R_{xx00} = 52500$



Sheet: /OUTPUT_CHANNELS/DCDC_2/
File: DCDC_ADJUSTABLE.kicad_sch

Title: *EPS*

Size: A4	Date: 2026-05-23
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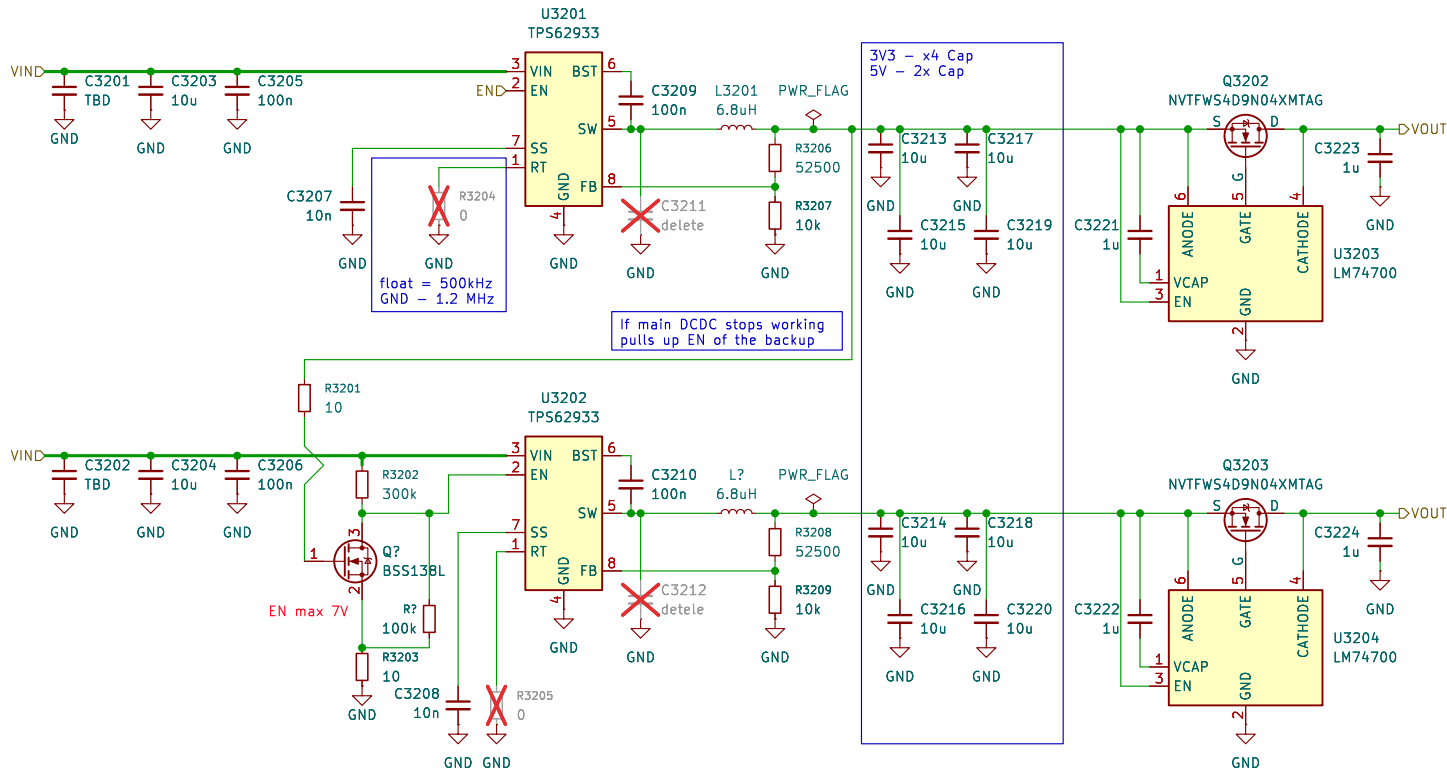
KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 40/54

Inputs	Outputs
BUS Voltage	Set voltage

Output voltage
5V
 $10000 \cdot (5V - 0.8) / 0.8 \Rightarrow R_{xx03}, R_{xx00} = 52500$



Sheet: /OUTPUT_CHANNELS/DCDC_3/
File: DCDC_ADJUSTABLE.kicad_sch

Title: EPS

Size: A4 Date: 2026-05-23

KiCad E.D.A. 9.0.4

Rev: Rev.1

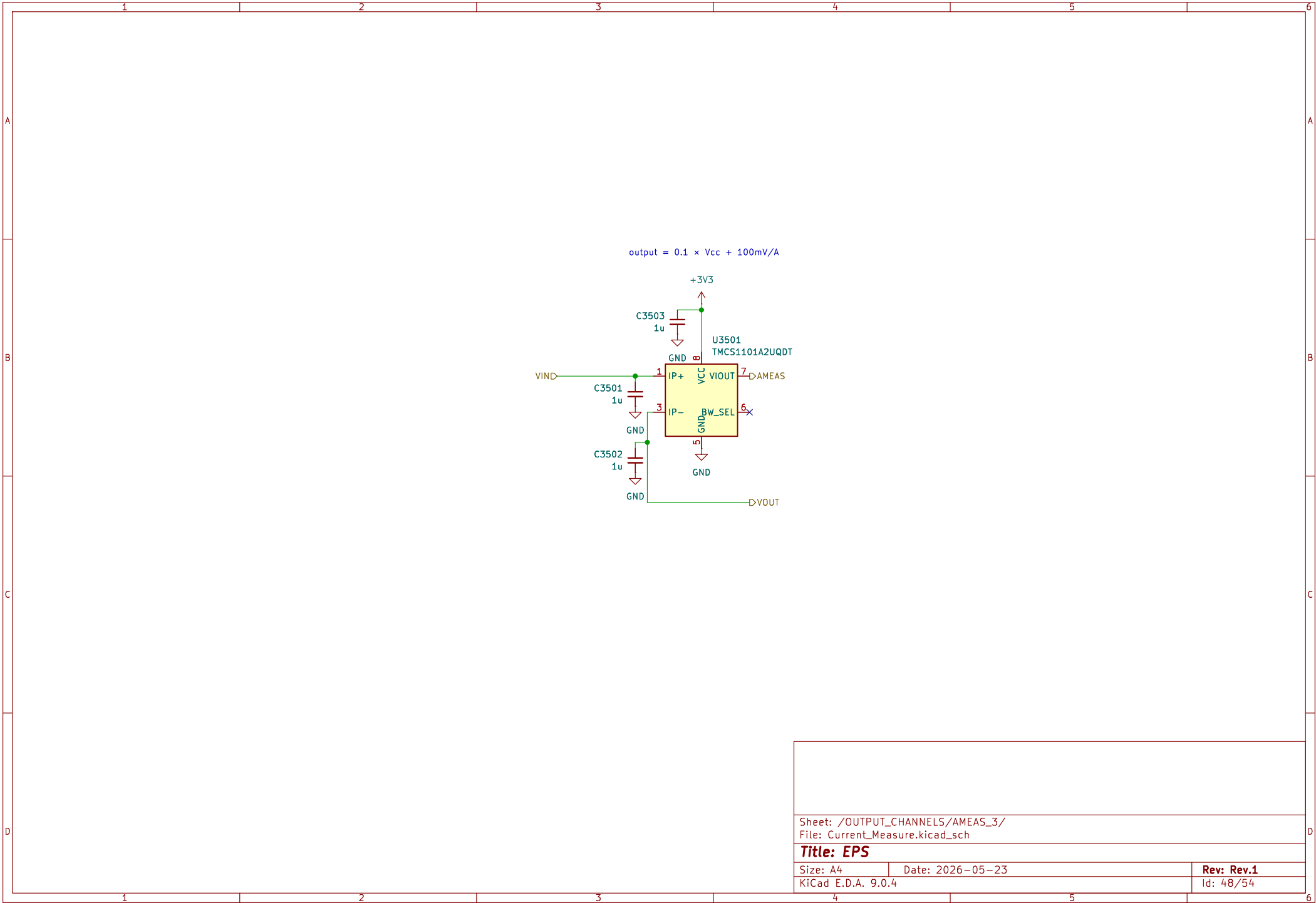
Id: 42/54



Sheet: /OUTPUT_CHANNELS/AMEAS_1/		
File: Current_Measure.kicad_sch		
Title: EPS		
Size: A4	Date: 2026-05-23	Rev: Rev.1
KiCad E.D.A. 9.0.4		Id: 43/54



Title: *EPS*



Sheet: /OUTPUT_CHANNELS/AMEAS_3/
File: Current_Measure.kicad_sch

Title: EPS

Size: A4 Date: 2026-05-23

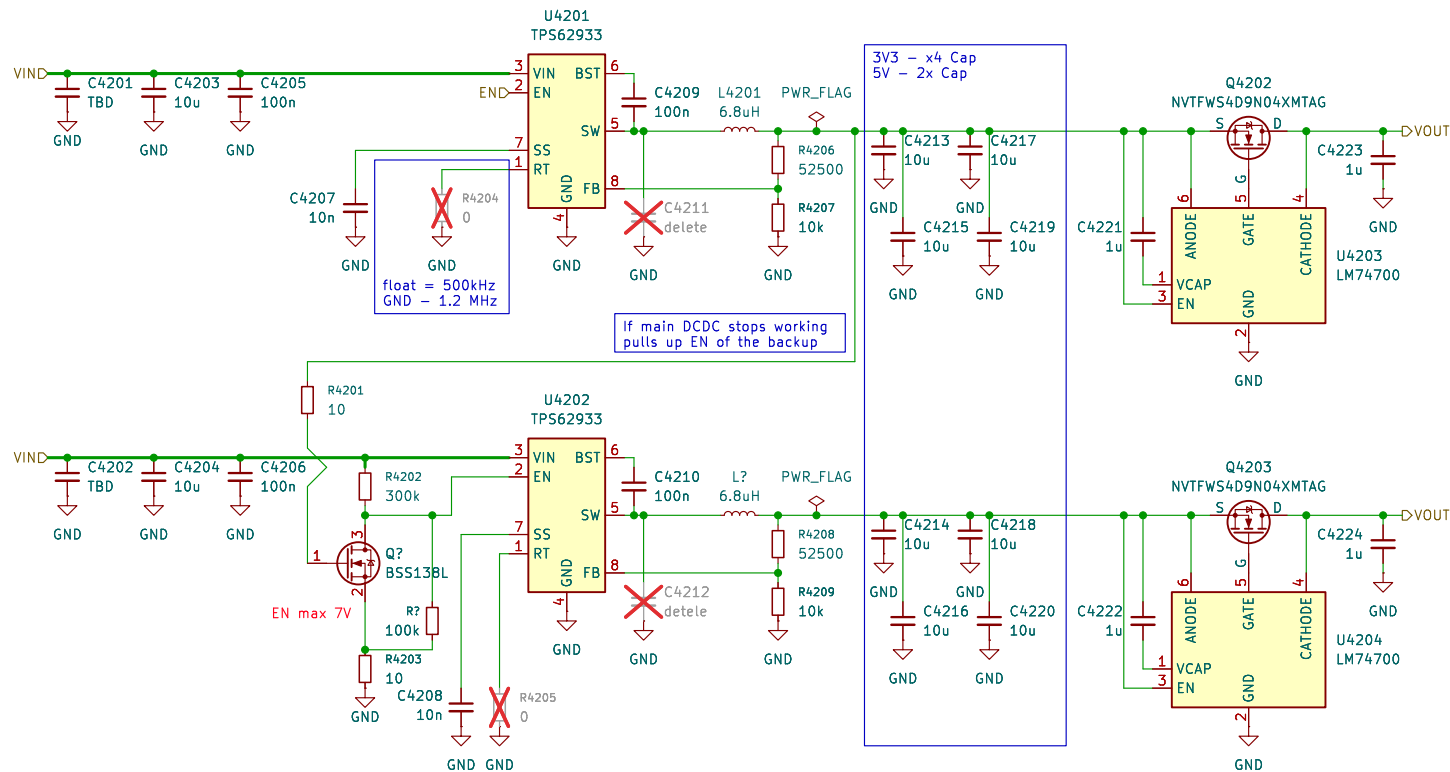
KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 48/54

Inputs	Outputs
BUS Voltage	Set voltage

Output voltage
5V
 $10000 \cdot (5V - 0,8) / 0,8 \Rightarrow R_{xx03}, R_{xx00} = 52500$



Sheet: /OUTPUT_CHANNELS/DCDC_7/
File: DCDC_ADJUSTABLE.kicad_sch

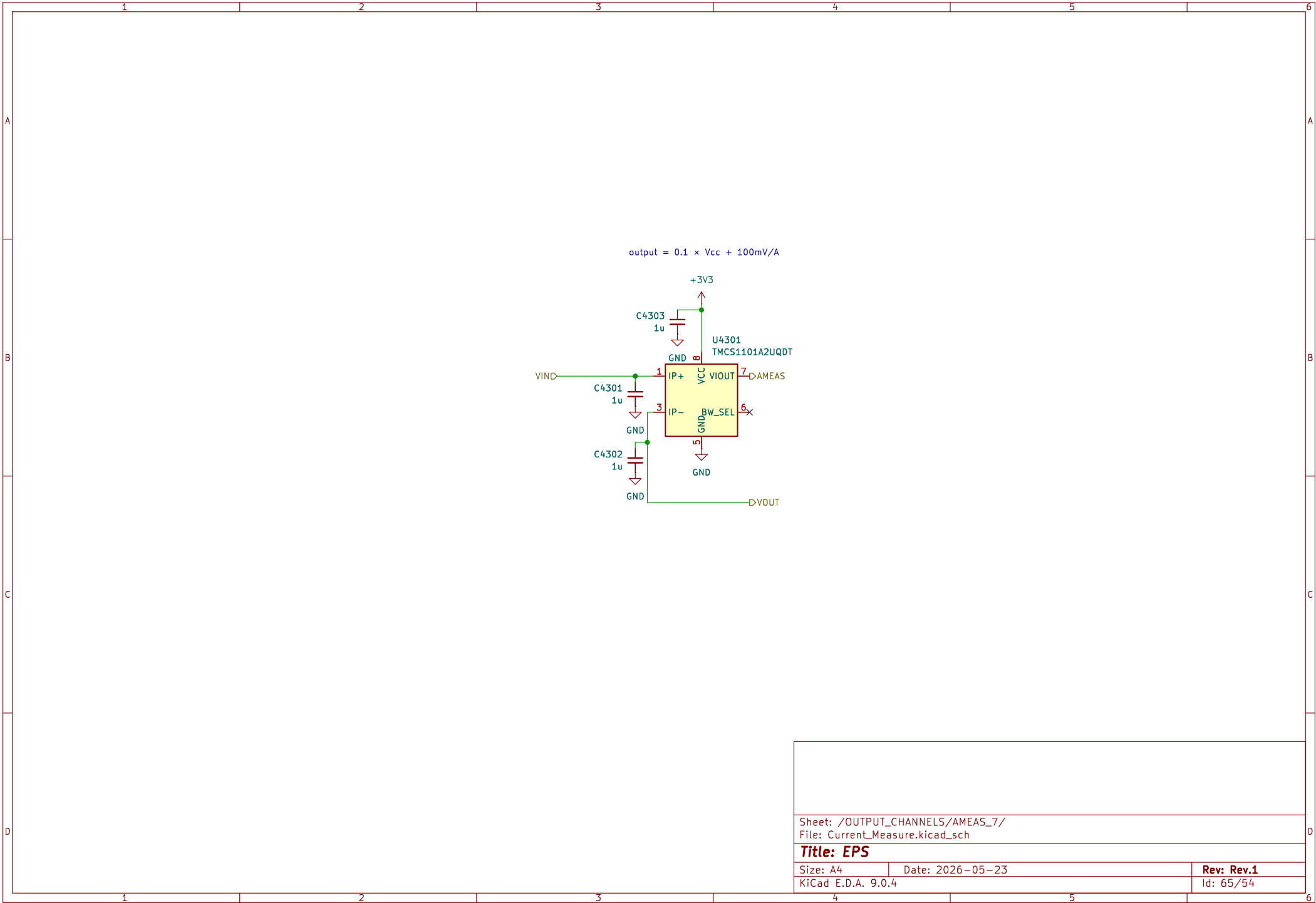
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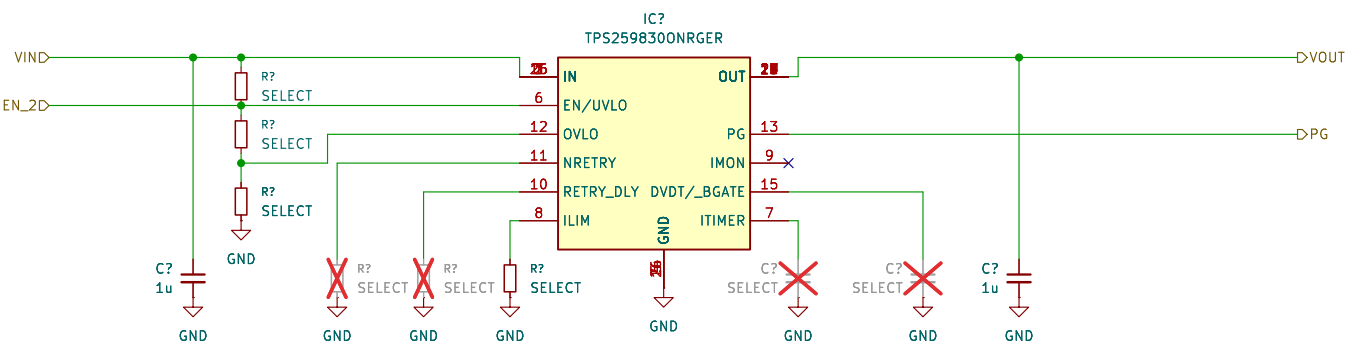
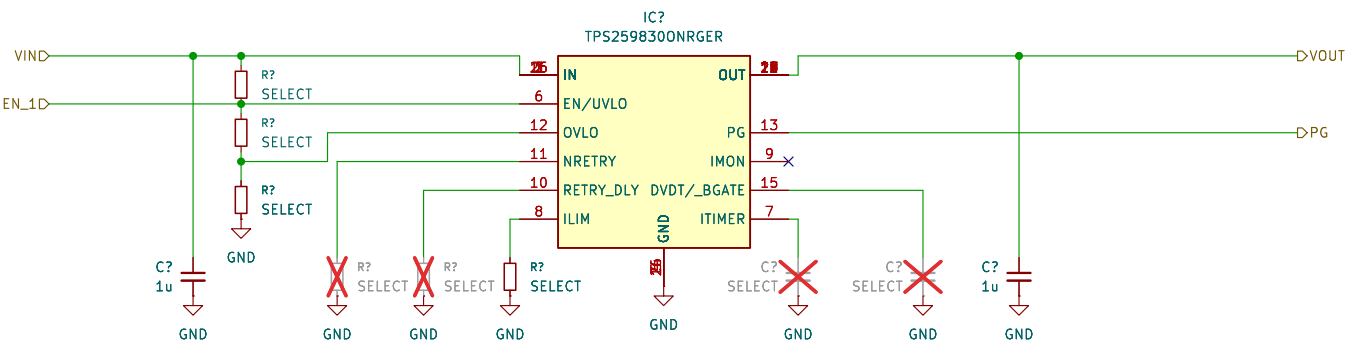
Size: A4	Date: 2026-05-23
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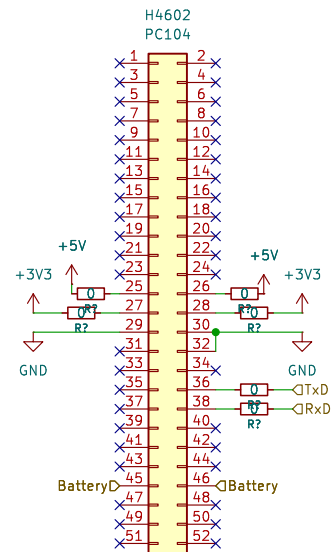
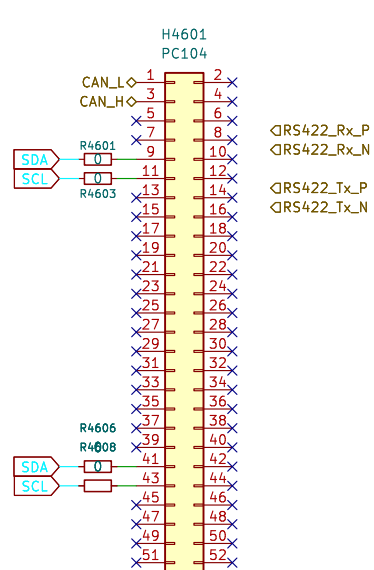
KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 61/54







Sheet: /OUTPUT_CHANNELS/PC104/
File: PC104.kicad_sch

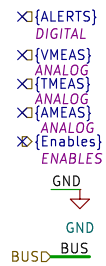
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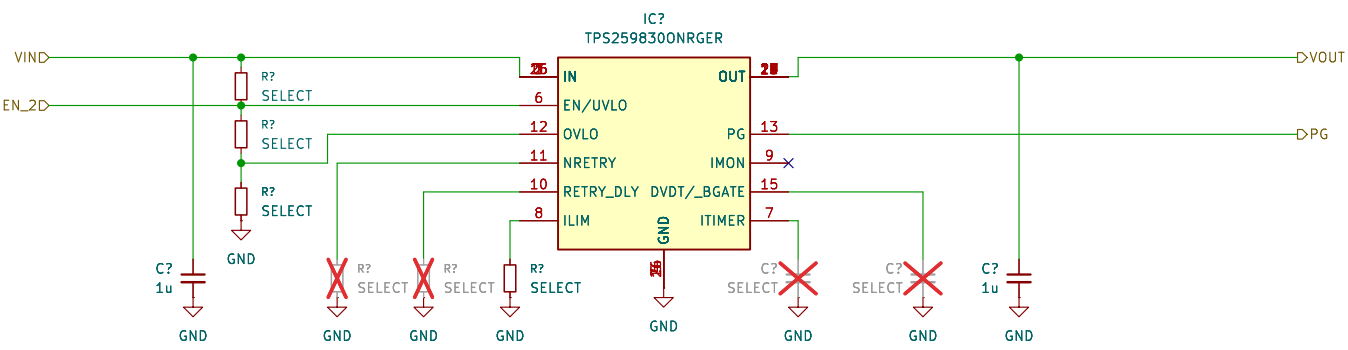
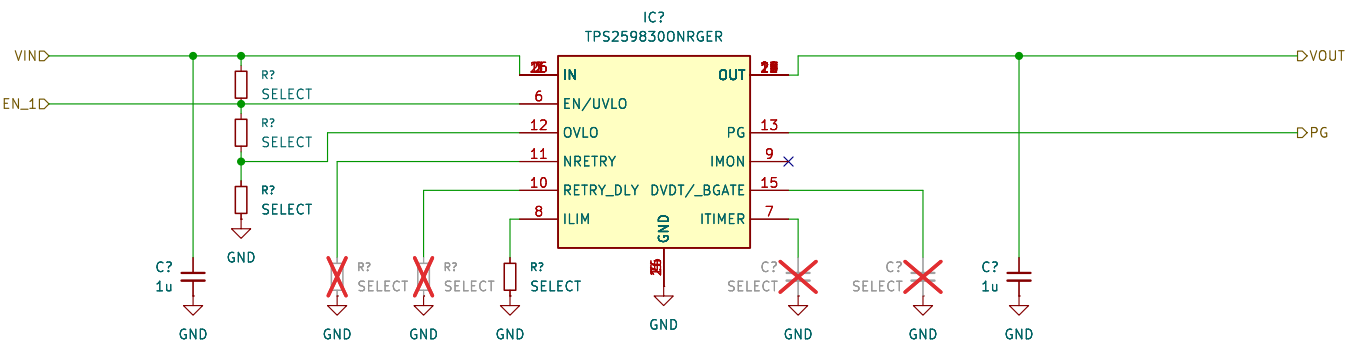
Size: A4 Date: 2026-05-23

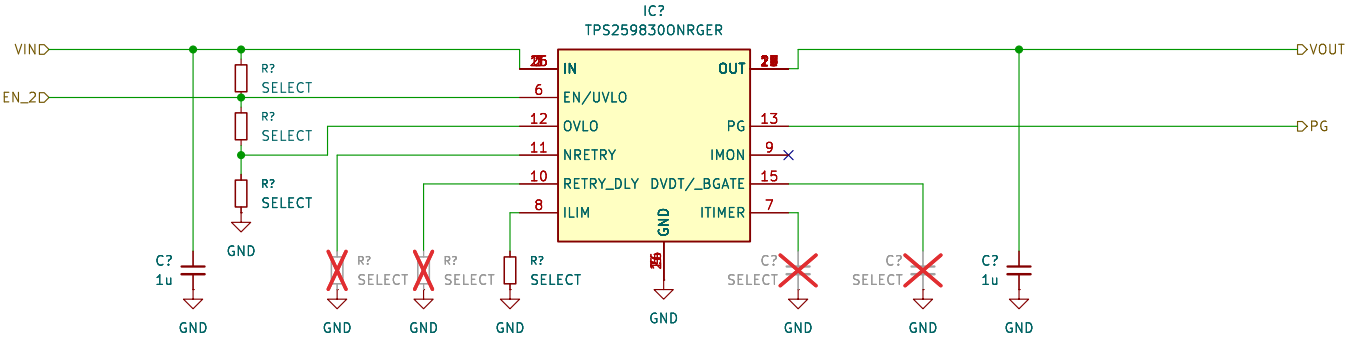
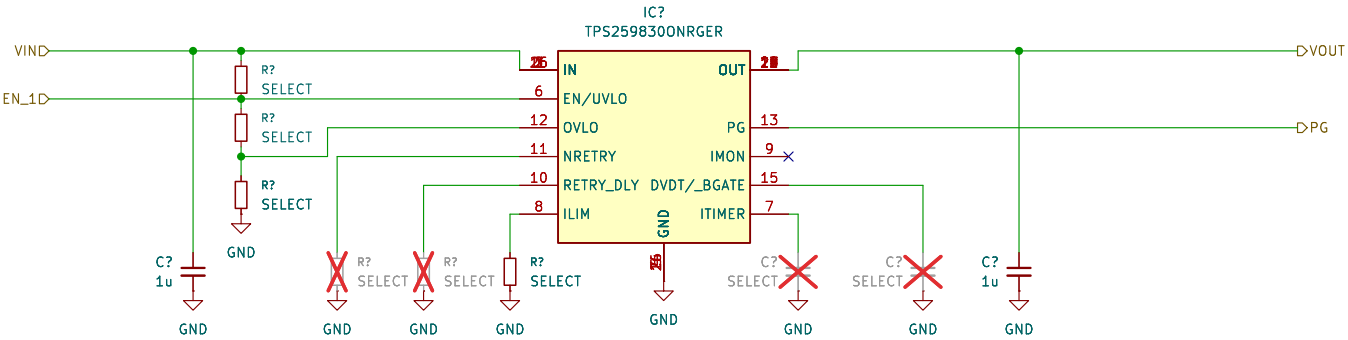
KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 70/54







Sheet: /OUTPUT_CHANNELS/Switch_A_2/
File: Switch_A.kicad_sch

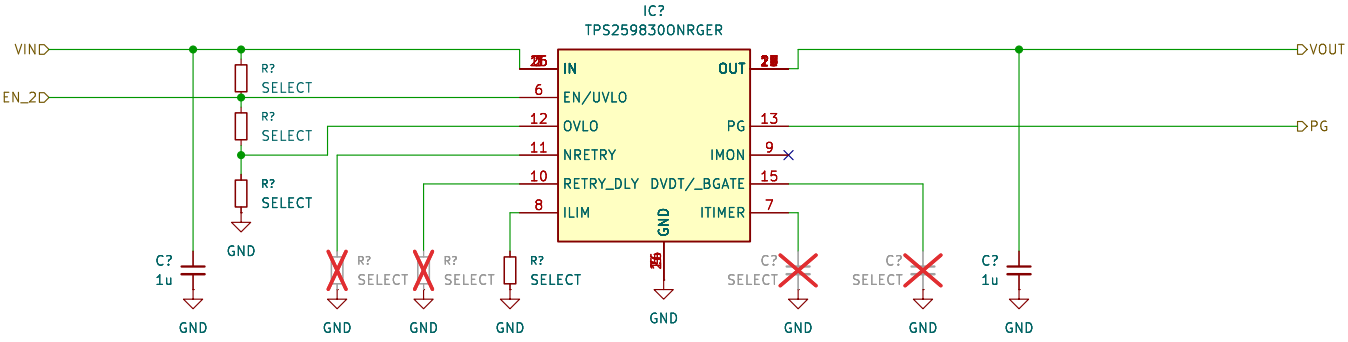
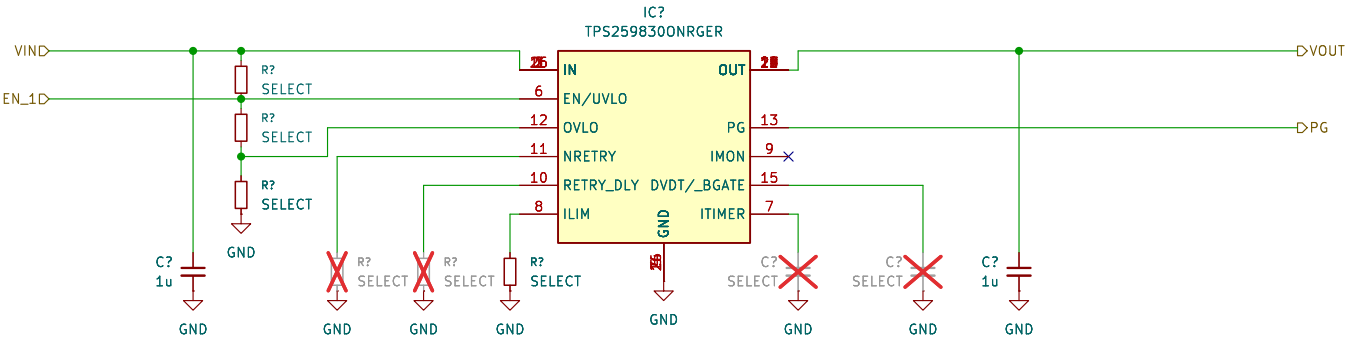
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Size: A4 Date: 2026-05-23

KiCad E.D.A. 9.0.4

Rev: Rev.1

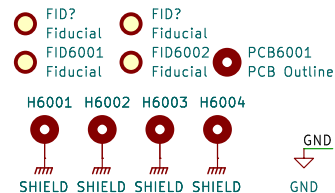
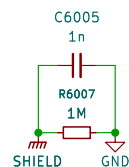
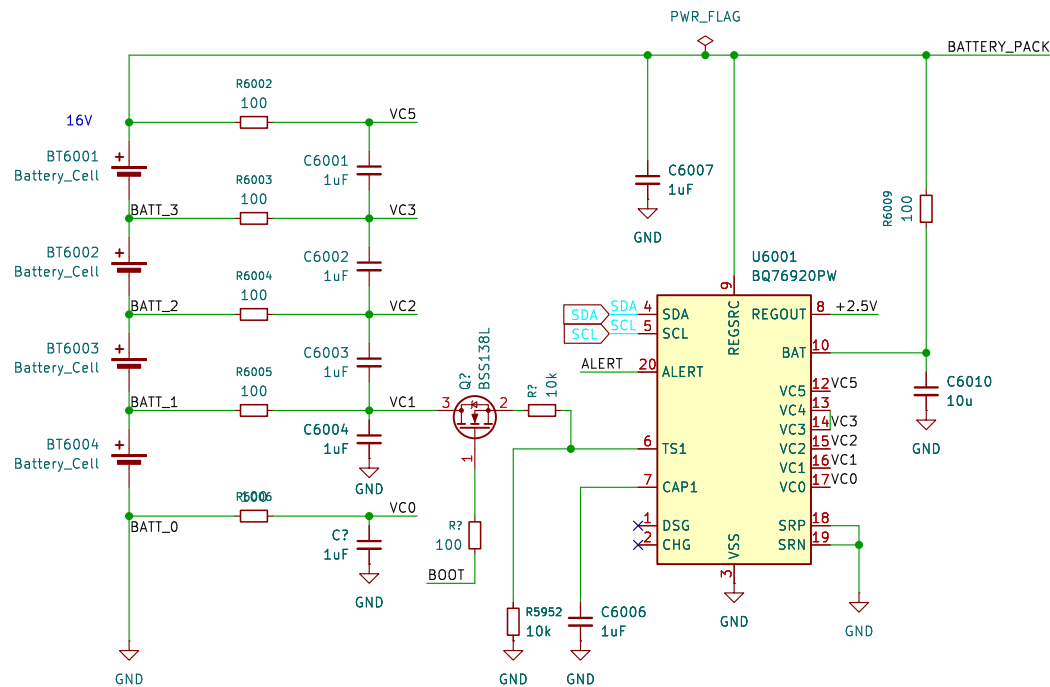
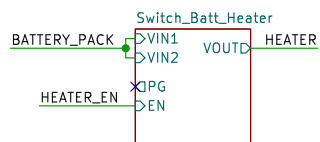
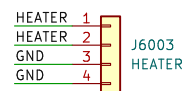
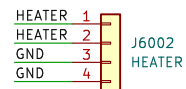
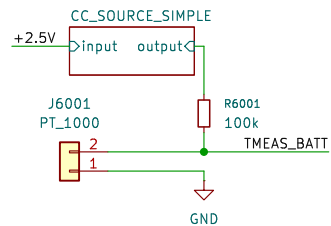
Id: 48/54



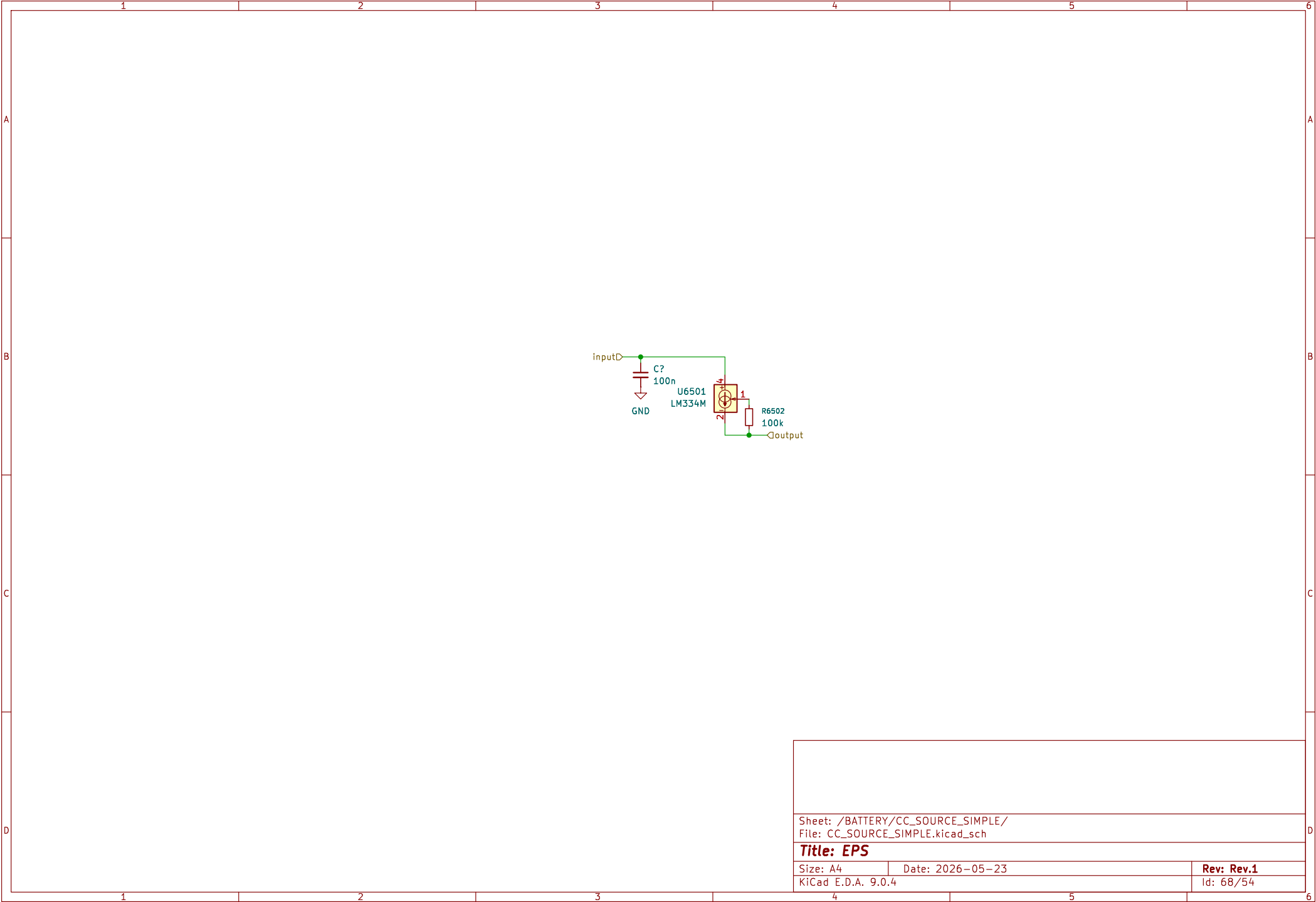
Inputs	Outputs
Battery	Battery voltage
Heater Enable	Measured voltage
4S battery	Temperature of the Cell
I2C	ALERT
BOOT	

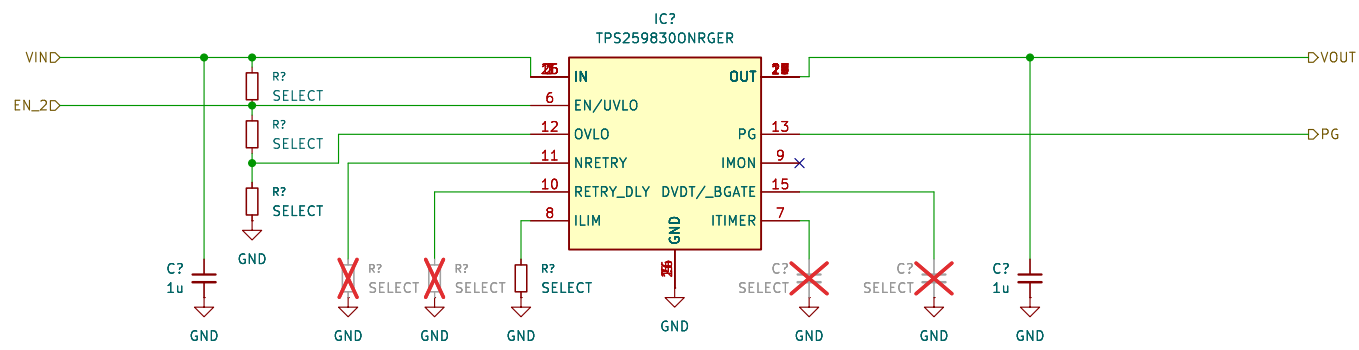
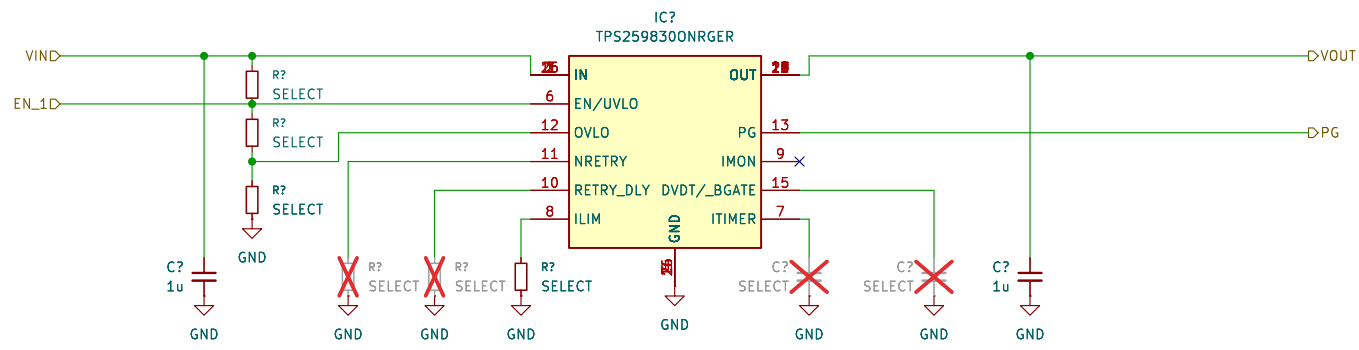
Input for Battery cell with
 -balancing circuit
 -Temperature measurement of the battery
 -Heater for maintaininig battery temperature

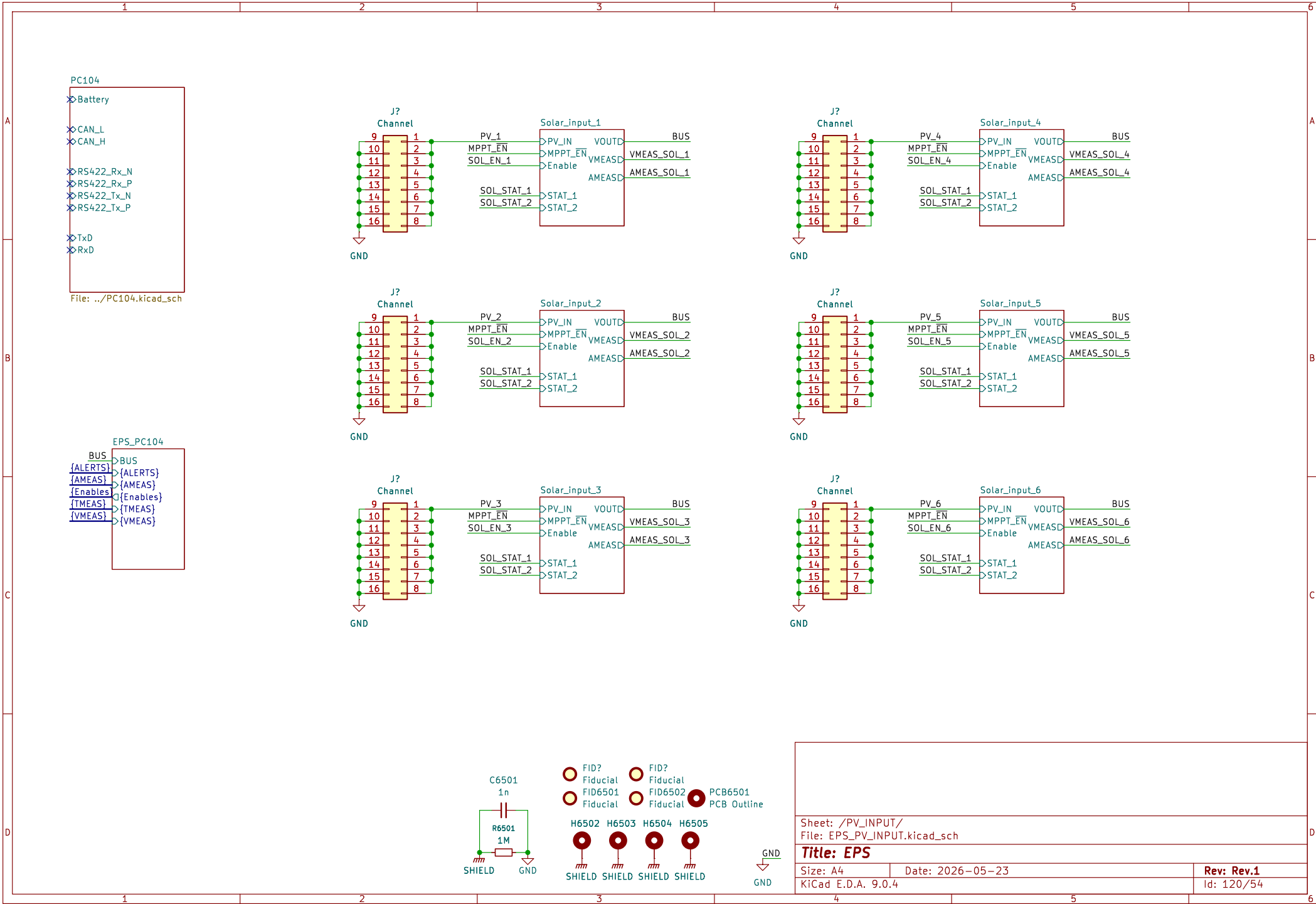
J6004		Battery	
BATTERY_PACK	1	9	BATTERY_PACK
BATTERY_PACK	2	10	BATTERY_PACK
GND	3	11	GND
TMEAS_BATT	4	12	GND
ALERT	5	13	GND
SCL	6	14	GND
SDA	7	15	GND
HEATER_EN	8	16	BOOT



Sheet: /BATTERY/	
File: EPS_BATTERY_PACK.kicad_sch	
Title: EPS	
Size: A4	Date: 2026-05-23
KiCad E.D.A. 9.0.4	Rev: Rev.1
Id: 106/54	





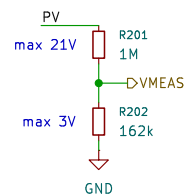
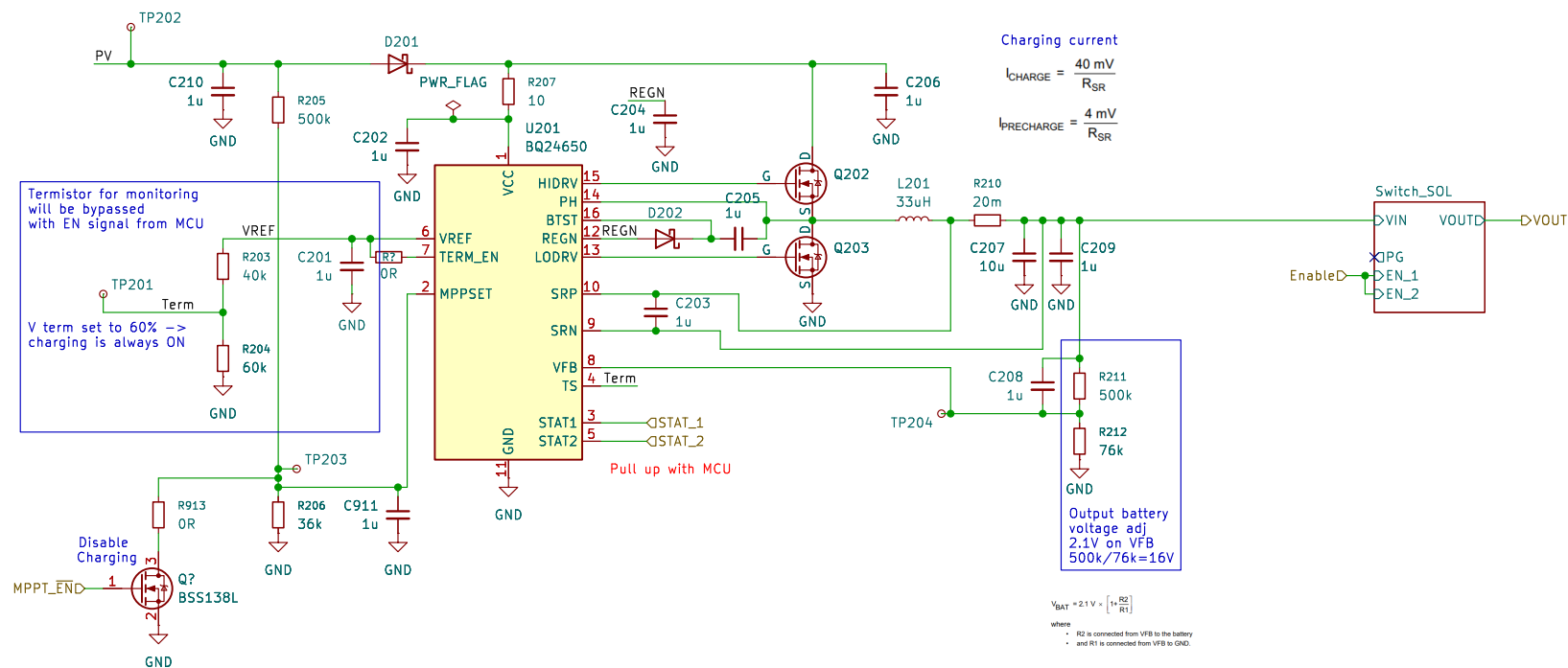
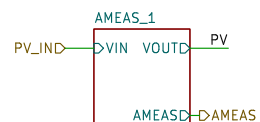


Inputs	Outputs
Solar Cell	Output voltage
PWM	Measured voltage
	Measured current

Input for solar cell with MPPT tracking
reverse current protection.

DC/DC controlled using PWM from CPU
to create MPPT function

```
solar voltage > VBUS
```



$$I_{\text{RIPPLE}} = \frac{V_{\text{IN}} \times D \times (1-D)}{f_s \times L}$$

$$(D = V_{OUT}/V_{IN}).$$

MPPTSET is regulated at 1.2V

$$V_{MPPSET} = 1.2 \text{ V} \times \left[1 + \frac{R3}{R4} \right]$$

Sheet: /PV_INPUT/Solar_input_2/
File: Solar_Cell_Input.kicad_sch

Title: EPS

Size: A4	Date: 2026-05-23
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KiCad E.D.A. 9.0.4

Rev: Rev.1

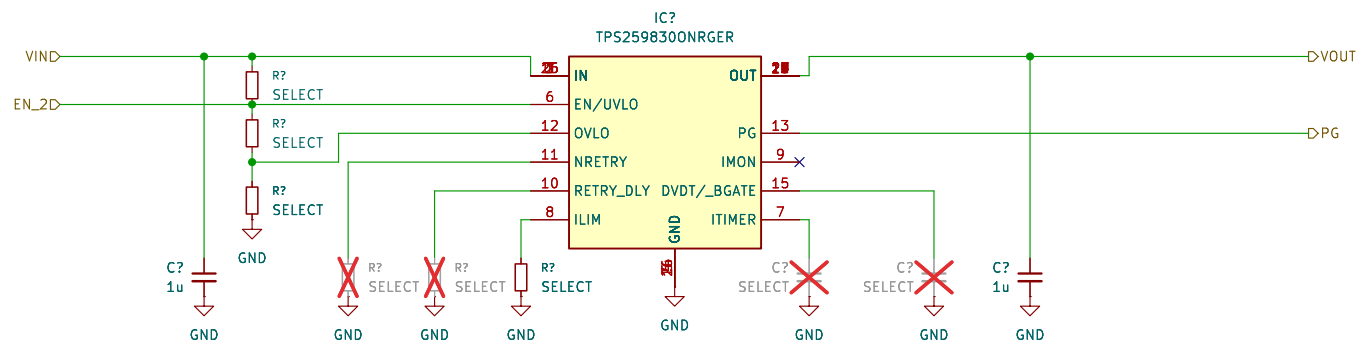
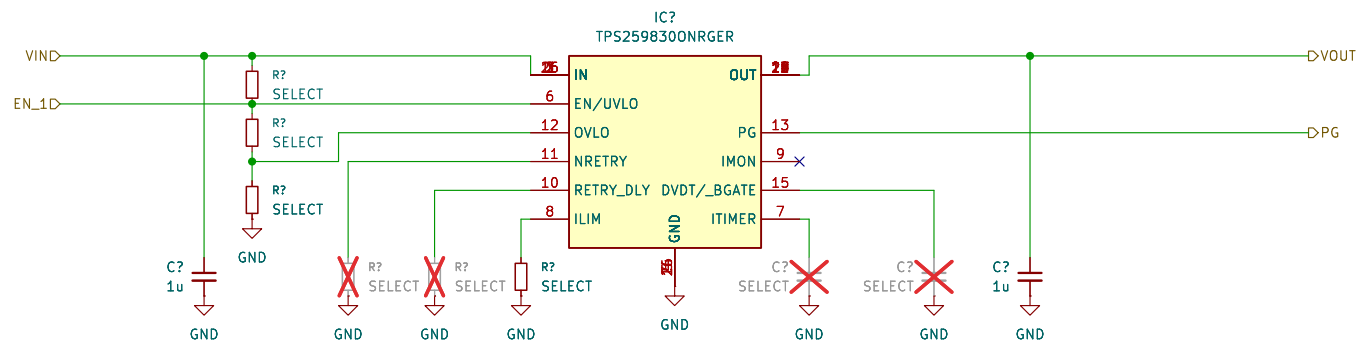
Id: 2/54



Sheet: /PV_INPUT/Solar_input_2/AMEAS_1/
File: Current_Measure.kicad_sch

Title: EPS

Size: A4	Date: 2026-05-23	Rev: Rev.1
KiCad E.D.A. 9.0.4		Id: 6/54



Sheet: /PV_INPUT/Solar_input_2/Switch_SOL/
File: Switch_A.kicad_sch

Title: *EPS*

Size: A4	Date: 2026-05-23
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Size: A4	
KiCad E.D.A. 9.0.4	

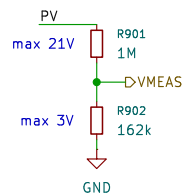
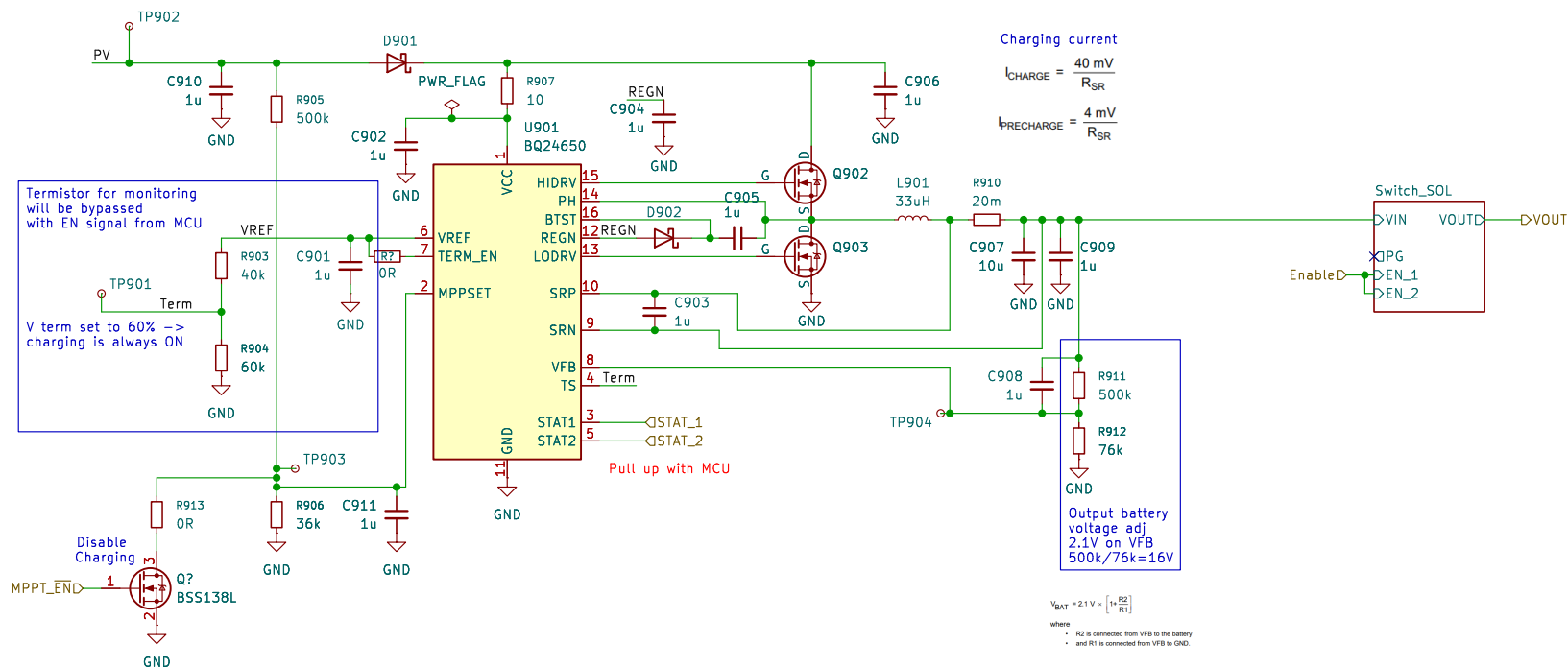
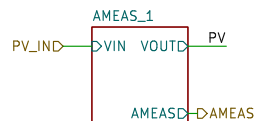
Rev: Rev.1

Id: 7/54

Input for solar cell with MPPT tracking
reverse current protection.

DC/DC controlled using PWM from CPU
to create MPPT function

solar voltage > VBUS



$$I_{\text{RIPPLE}} = \frac{V_{\text{IN}} \times D \times (1-D)}{f_s \times L}$$

$$(D = V_{OUT}/V_{IN}).$$

MPPTSET is regulated at 1.2V

$$V_{MPPSET} = 1.2 \text{ V} \times \left[1 + \frac{R3}{R4} \right]$$

Sheet: /PV_INPUT/Solar_input_3/
File: Solar_Cell_Input.kicad_sch

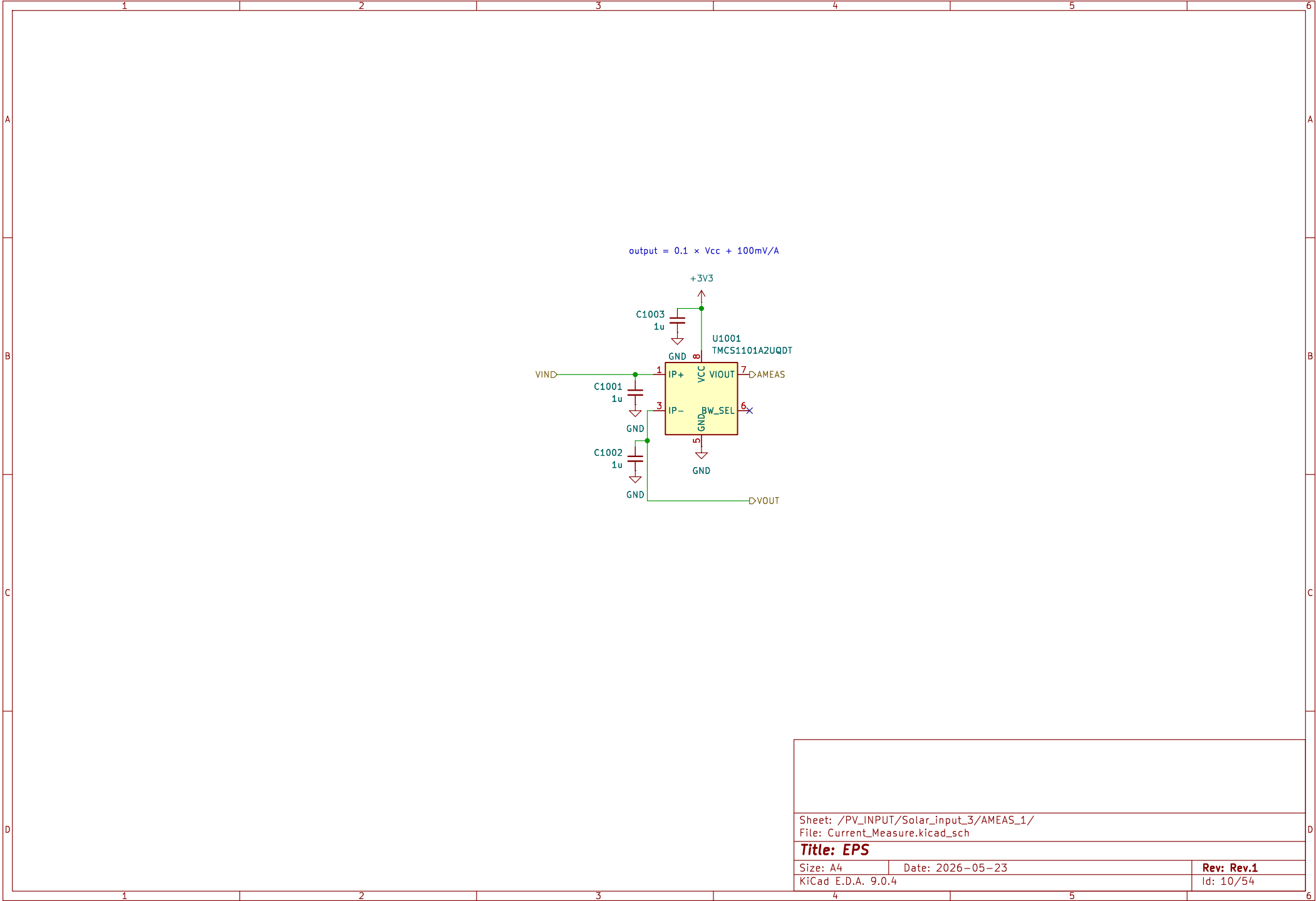
Title: EPS

Size: A4	Date: 2026-05-23	Rev: Rev.1
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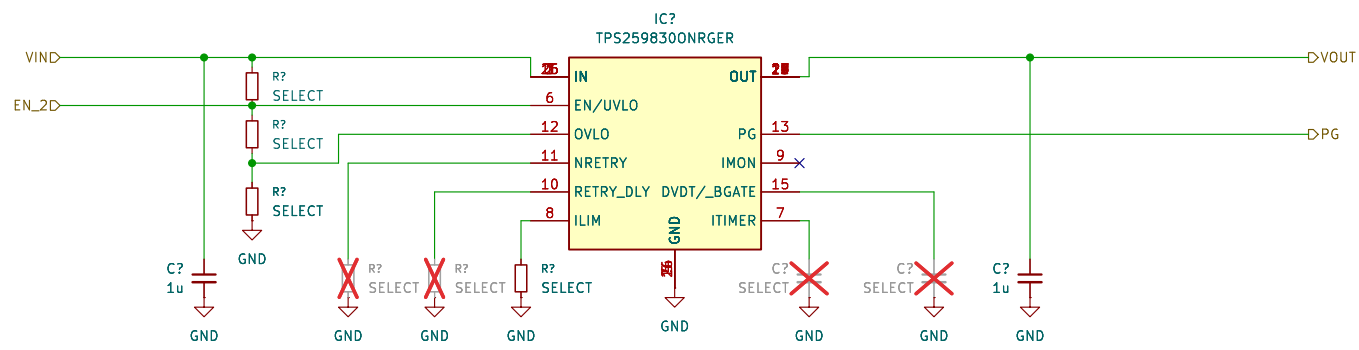
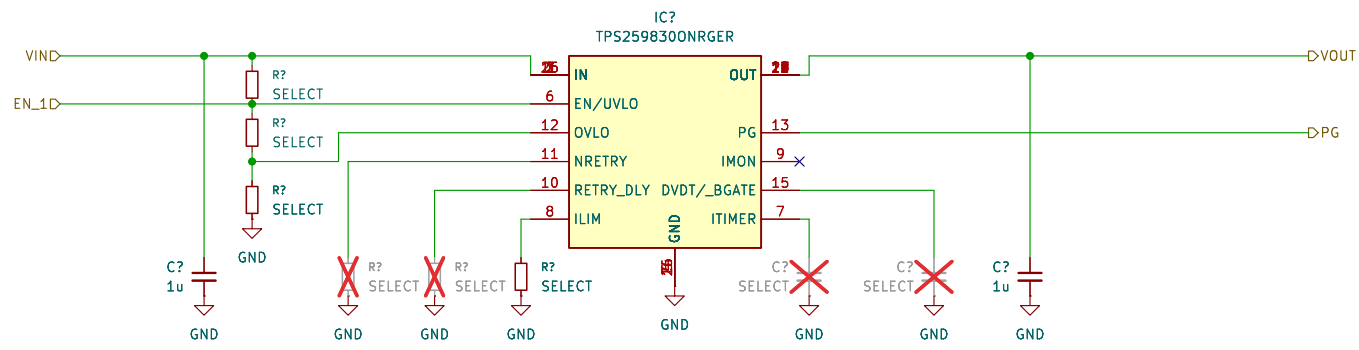
Rev: Rev1	Date: 2020-05-25	Rev: Rev1
KiCad E.D.A. 9.0.4		Id: 9/54

Rev: Rev.1

Id: 9/54



Sheet: /PV_INPUT/Solar_input_3/AMEAS_1/		
File: Current_Measure.kicad_sch		
Title: EPS		
Size: A4	Date: 2026-05-23	Rev: Rev.1
KiCad E.D.A. 9.0.4	Id: 10/54	



Sheet: /PV_INPUT/Solar_input_3/Switch_SOL/
File: Switch_A.kicad_sch

Title: EPS

Size: A4 Date: 2026-05-23

KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 11/54

Input for solar cell with MPPT tracking
reverse current protection.

DC/DC controlled using PWM from CPU
to create MPPT function

TP1302

PV

C1310 1u

GND

R1305 500k

D1301

PWR_FLAG

R1307 10

U1301 BQ24650

REGN

C1304 1u

GND

C1306 1u

GND

Charging current

$$I_{CHARGE} = \frac{40 \text{ mV}}{R_{SR}}$$

$$I_{PRECHARGE} = \frac{4 \text{ mV}}{R_{SR}}$$

Termistor for monitoring will be bypassed with EN signal from MCU

TP1301

Term

V term set to 60% -> charging is always ON

VREF

R1303 40k

C1301 1u

GND

R1304 60k

GND

TP1303

R1306 36k

C911 1u

GND

R913 0R

MPPT_END

Q1 BSS138L

GND

VCC

HIDRV

PH

BTST

REGN

LODRV

SRP

SRN

VFB

TS

STAT1

STAT2

GND

Q1302

Q1303

D1302

C1305 1u

GND

L1301 33uH

R1310 20m

C1307 10u

GND

C1309 1u

GND

Switch_SOL

VIN

VOUT

Enable

EN_1

EN_2

TP1304

C1308 1u

GND

R1311 500k

R1312 76k

GND

Output battery voltage adj 2.1V on VFB 500k/76k=16V

Pull up with MCU

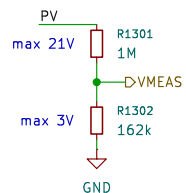
STAT_1

STAT_2

$V_{BAT} = 2.1 \text{ V} \times \left[1 + \frac{R2}{R1} \right]$

where

- R2 is connected from VFB to the battery
- and R1 is connected from VFB to GND.



$$(D = V_{OUT}/V_{IN})$$

MPPTSET is regulated at 1.2V

$$V_{MPPSET} = 1.2 \text{ V} \times \left[1 + \frac{R3}{R4} \right]$$

Sheet: /PV_INPUT/Solar_input_4/
File: Solar_Cell_Input.kicad_sch

Title: *EPS*

Size: A4	Date: 2026-05-23
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KiCad E.D.A. 9.0.4

Rev: Rev.1

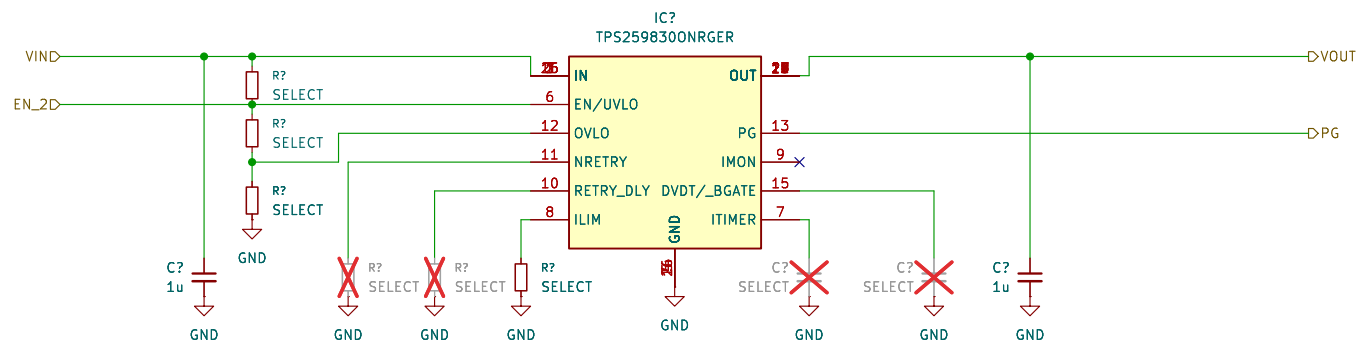
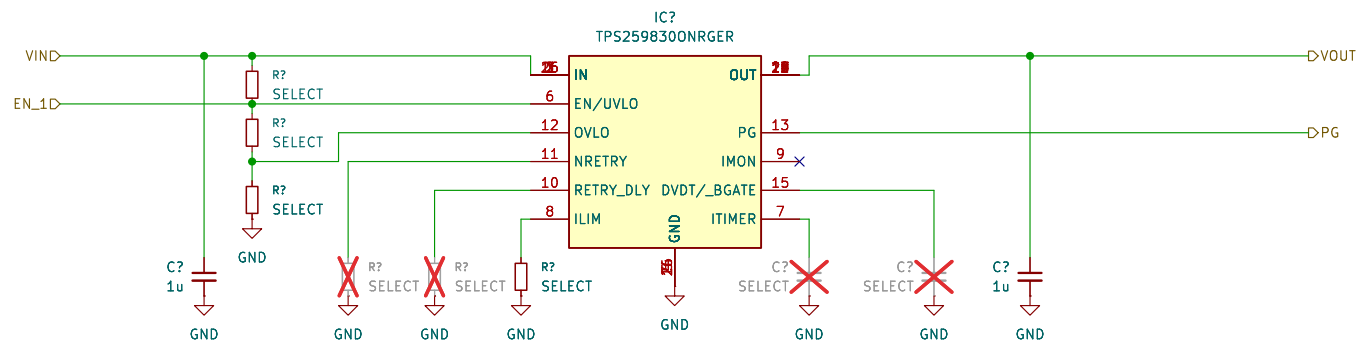
Id: 13/54



Title: EPS

Rev: Rev.1

Id: 14/54



Sheet: /PV_INPUT/Solar_input_4/Switch_SOL/
File: Switch_A.kicad_sch

Title: EPS

Size: A4	Date: 2026-05-23
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Size: A4	
KiCad E.D.A. 9.0.4	

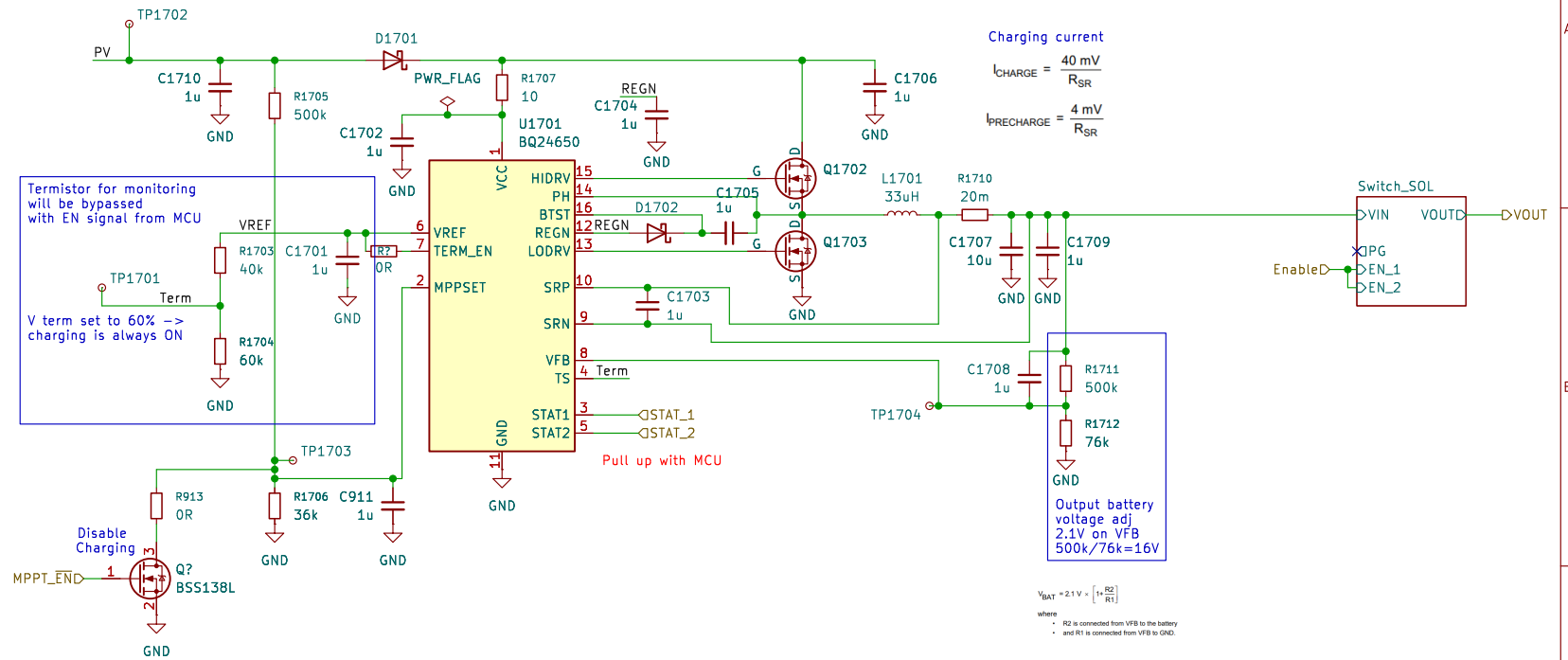
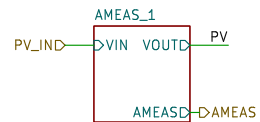
Rev: Rev.1

Id: 15/54

Inputs	Outputs
Solar Cell	Output voltage
PWM	Measured voltage
	Measured current

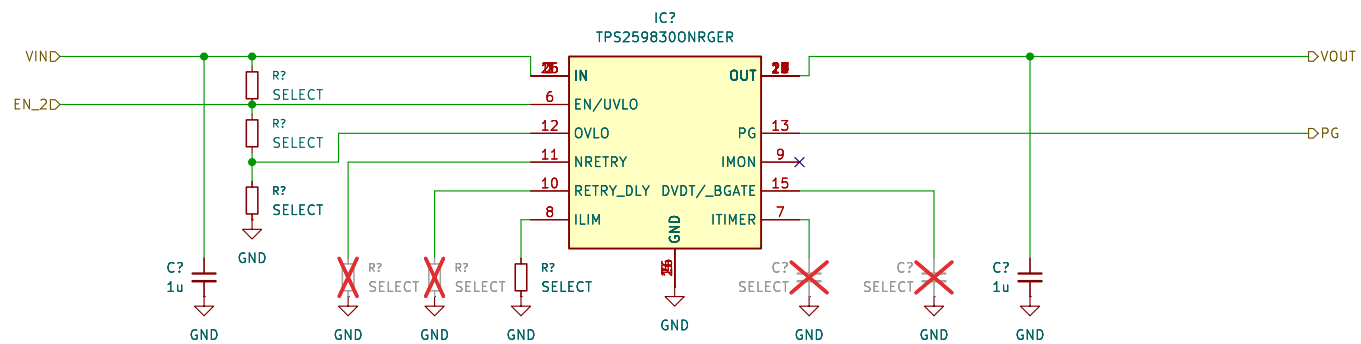
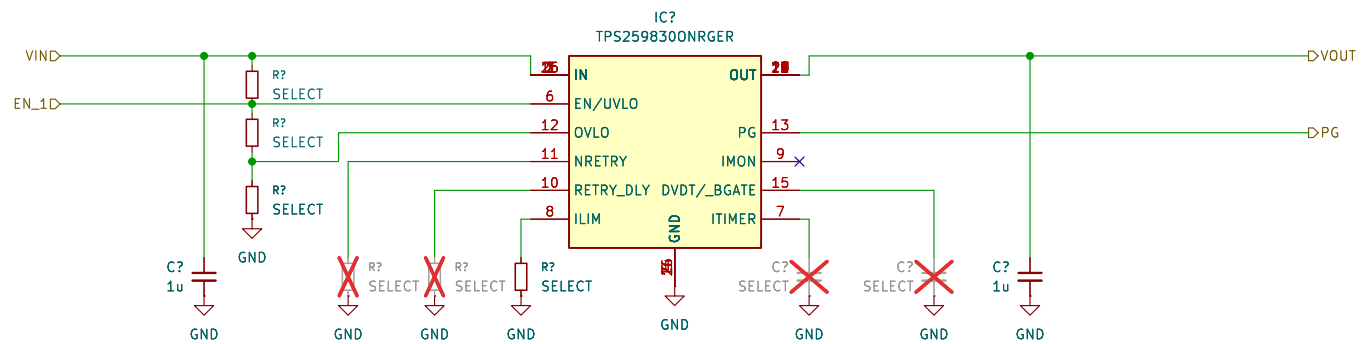
Input for solar cell with MPPT tracking reverse current protection.
DC/DC controlled using PWM from CPU to create MPPT function

solar voltage > VBUS





Sheet: /PV_INPUT/Solar_input_5/AMEAS_1/ File: Current_Measure.kicad_sch		D
Title: EPS		
Size: A4	Date: 2026-05-23	
KiCad E.D.A. 9.0.4	Rev: Rev.1 Id: 18/54	



Sheet: /PV_INPUT/Solar_input_5/Switch_SOL/
File: Switch_A.kicad_sch

Title: *EPS*

Size: A4	Date: 2026-05-23
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KiCad E.D.A. 9.0.4

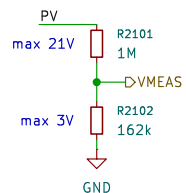
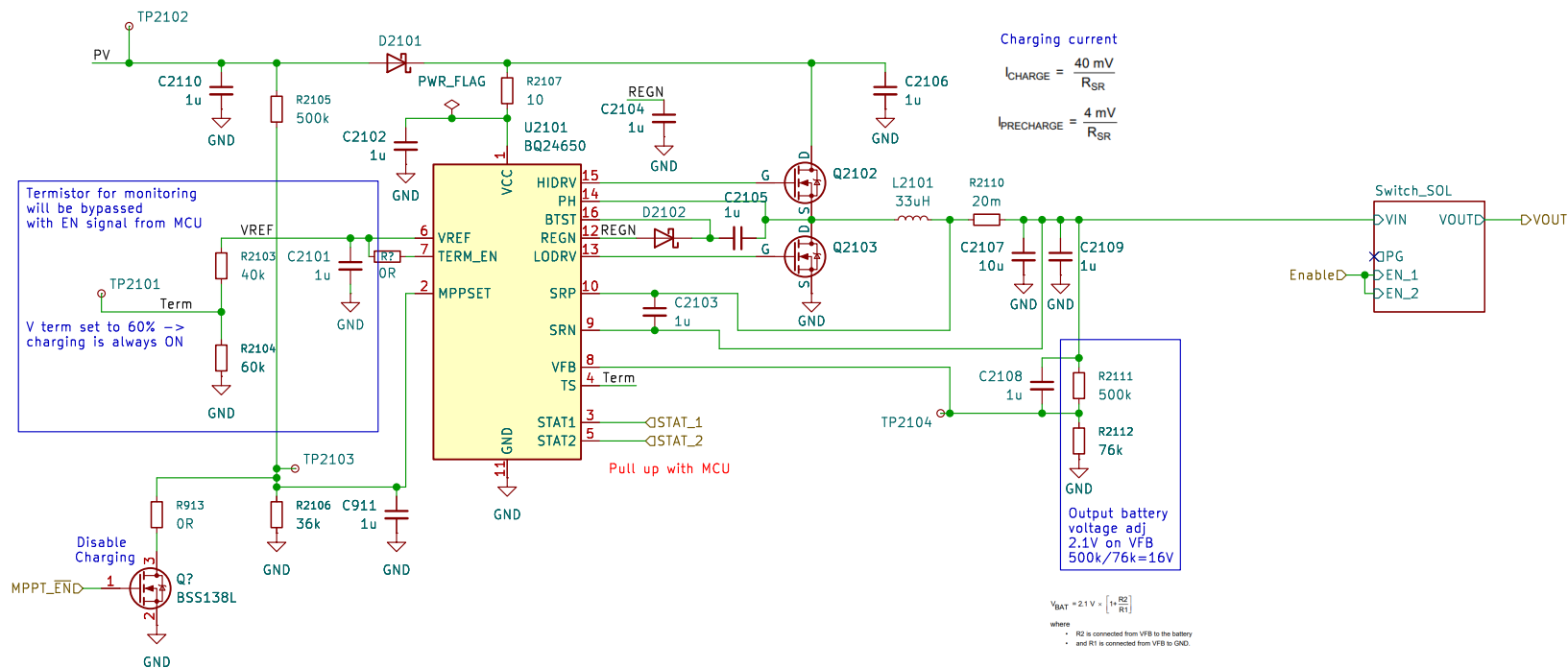
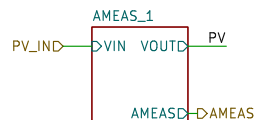
Rev: Rev.1

Id: 19/54

Input for solar cell with MPPT tracking
reverse current protection.

DC/DC controlled using PWM from CPU
to create MPPT function

solar voltage > VBUS



$$I_{\text{RIPPLE}} = \frac{V_{\text{IN}} \times D \times (1-D)}{f_s \times L}$$

$$(D = V_{OUT}/V_{IN}).$$

MPPTSET is regulated at 1.2V

$$V_{MPPSET} = 1.2 \text{ V} \times \left[1 + \frac{R3}{R4} \right]$$

Sheet: /PV_INPUT/Solar_input_6/
File: Solar_Cell_Input.kicad_sch

Title: *EPS*

Size: A4	Date: 2026-05-23
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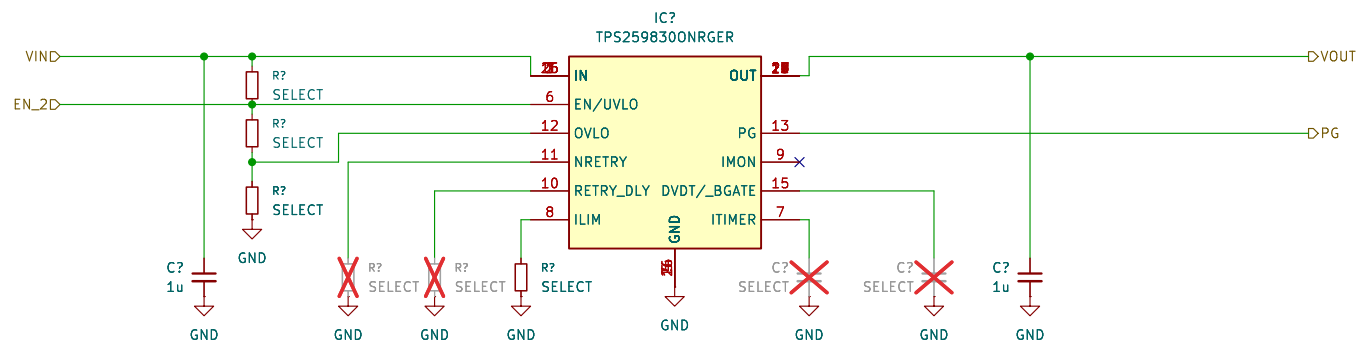
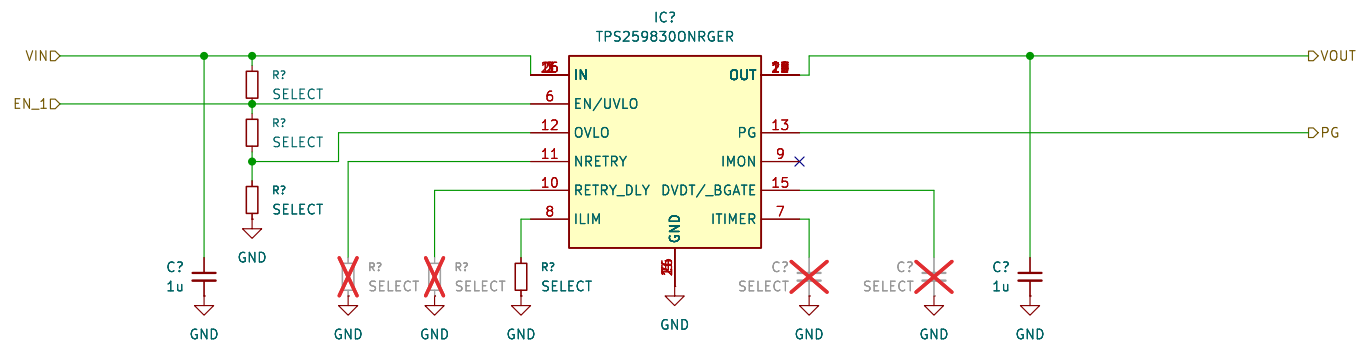
KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 21/54



Sheet: /PV_INPUT/Solar_input_6/AMEAS_1/		
File: Current_Measure.kicad_sch		
Title: EPS		
Size: A4	Date: 2026-05-23	Rev: Rev.1
KiCad E.D.A. 9.0.4		Id: 22/54



Sheet: /PV_INPUT/Solar_input_6/Switch_SOL/
File: Switch_A.kicad_sch

Title: *EPS*

Size: A4	Date: 2026-05-23
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Size: A4	
KiCad E.D.A. 9.0.4	

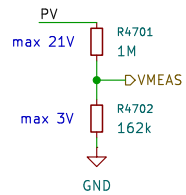
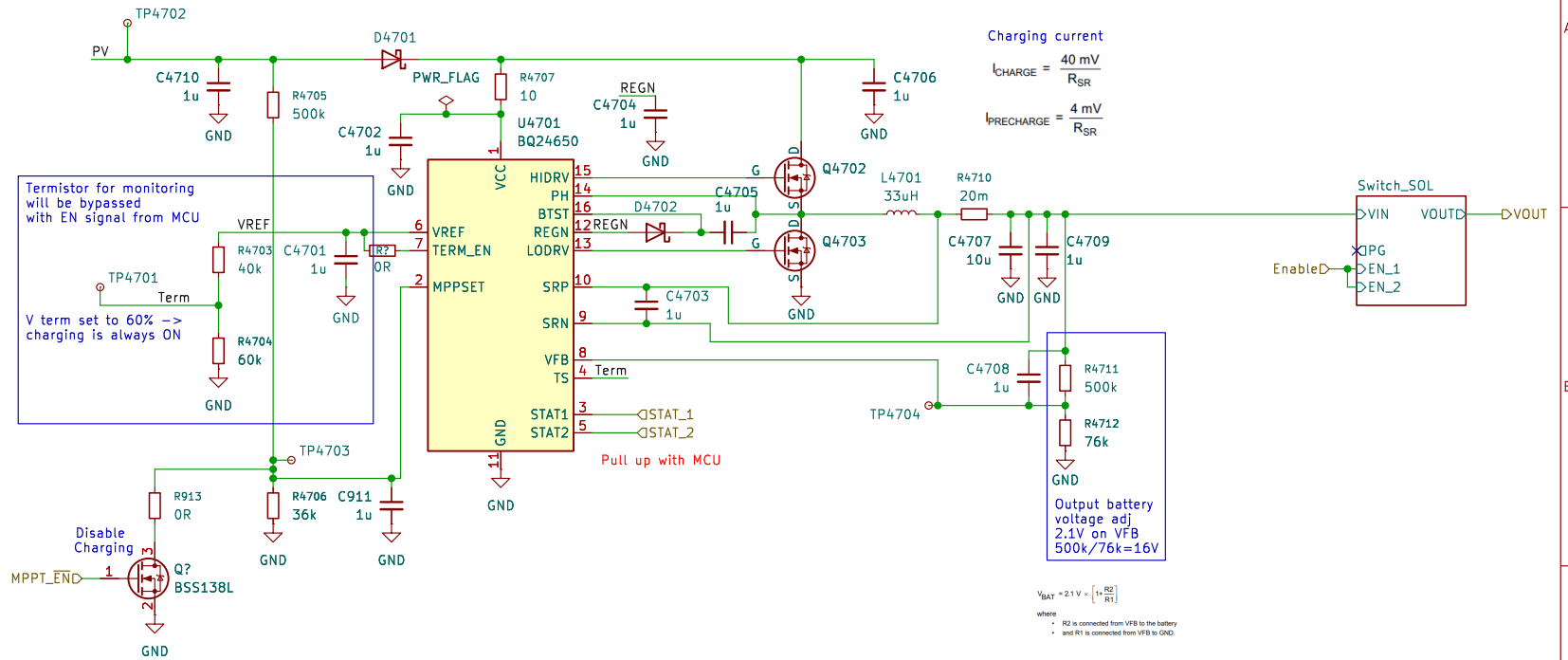
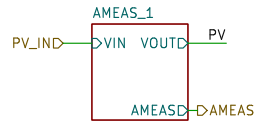
Rev: Rev.1

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Inputs	Outputs
Solar Cell	Output voltage
PWM	Measured voltage
	Measured current

Input for solar cell with MPPT tracking reverse current protection.
DC/DC controlled using PWM from CPU to create MPPT function

solar voltage > VBUS



$$I_{RIPPLE} = \frac{V_{IN} \times D \times (1-D)}{f_s \times L}$$

$$(D = V_{OUT}/V_{IN})$$

Sheet: /PV_INPUT/Solar_input_1/
File: Solar_CellInput.kicad_sch

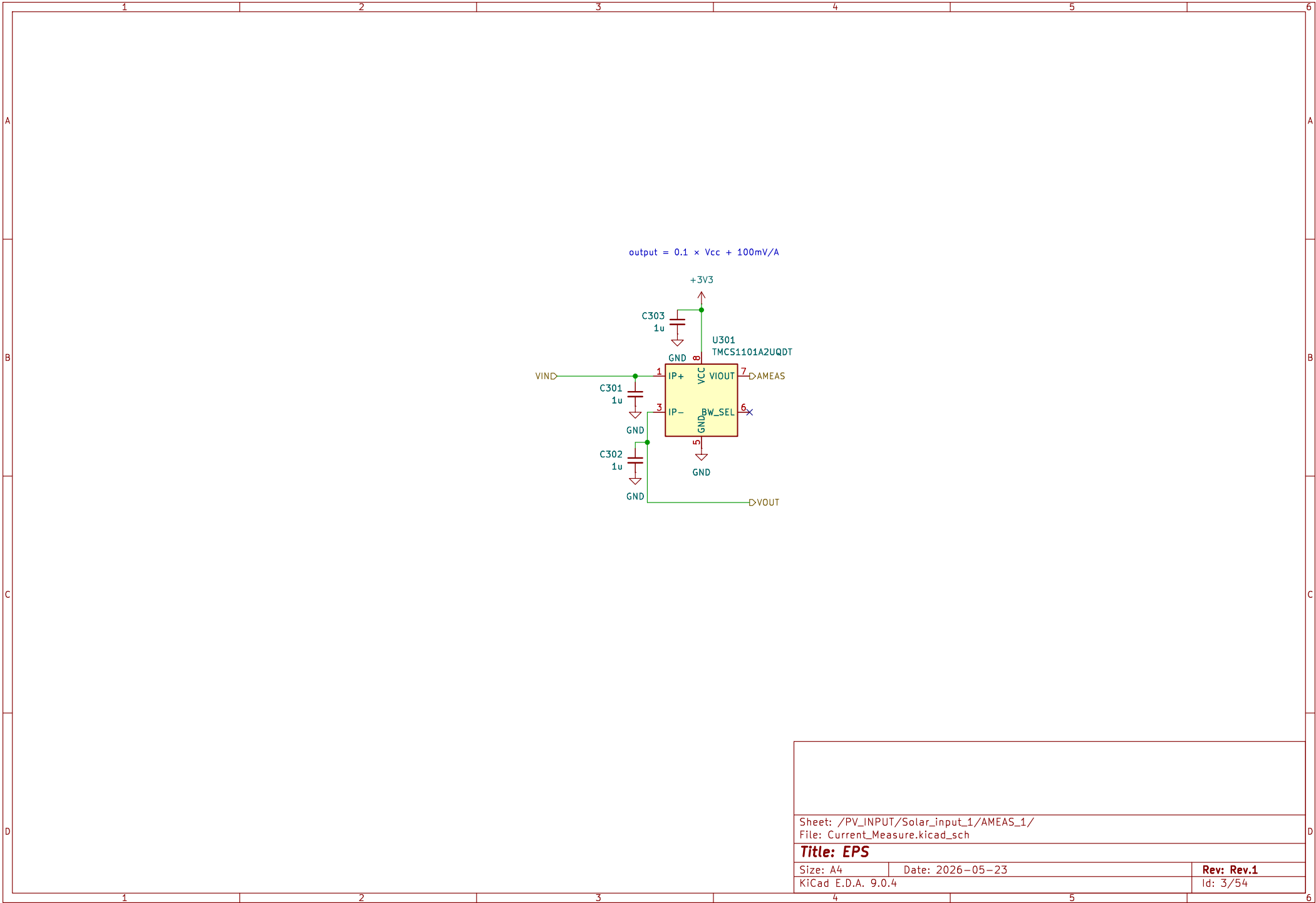
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Size: A4 Date: 2026-05-23

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Sheet: /PV_INPUT/Solar_input_1/AMEAS_1/
File: Current_Measure.kicad_sch

Title: EPS

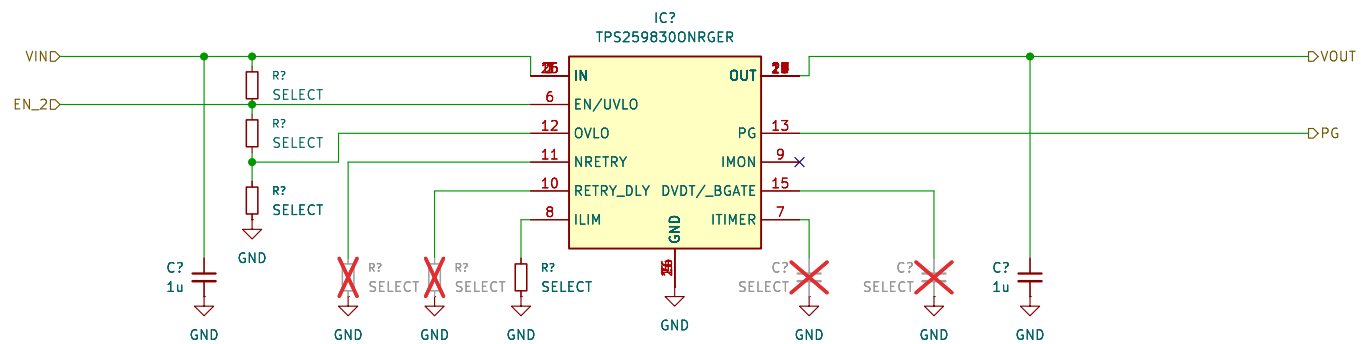
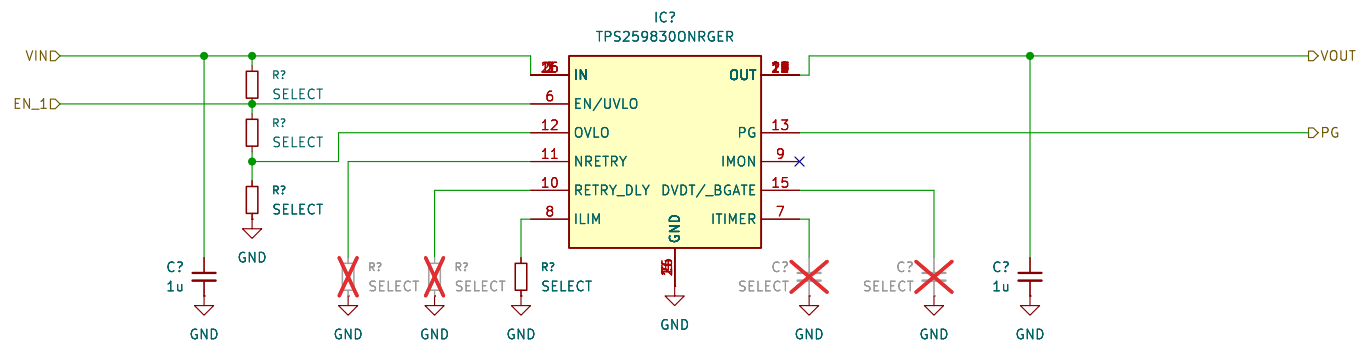
Size: A4

Date: 2026-05-23

Rev: Rev.1

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Sheet: /PV_INPUT/Solar_input_1/Switch_SOL/
File: Switch_A.kicad_sch

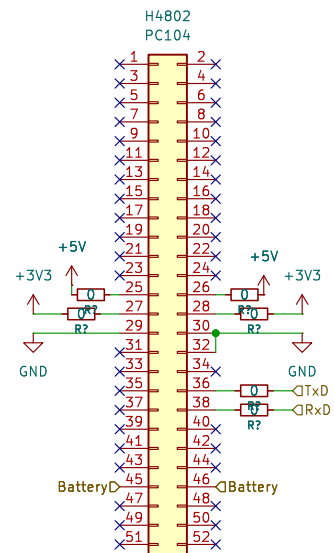
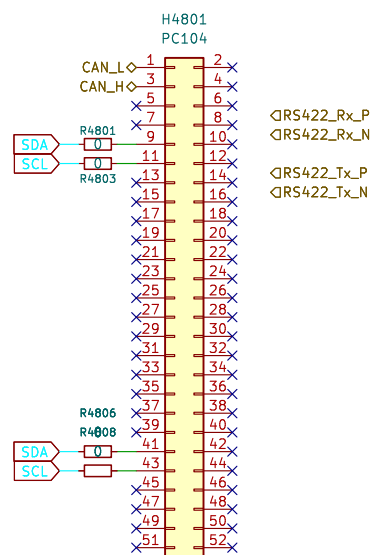
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Size: A4	Date: 2026-05-23
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Size: A4	
KiCad E.D.A. 9.0.4	

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Sheet: /PV_INPUT/PC104/
File: PC104.kicad_sch

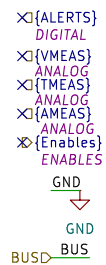
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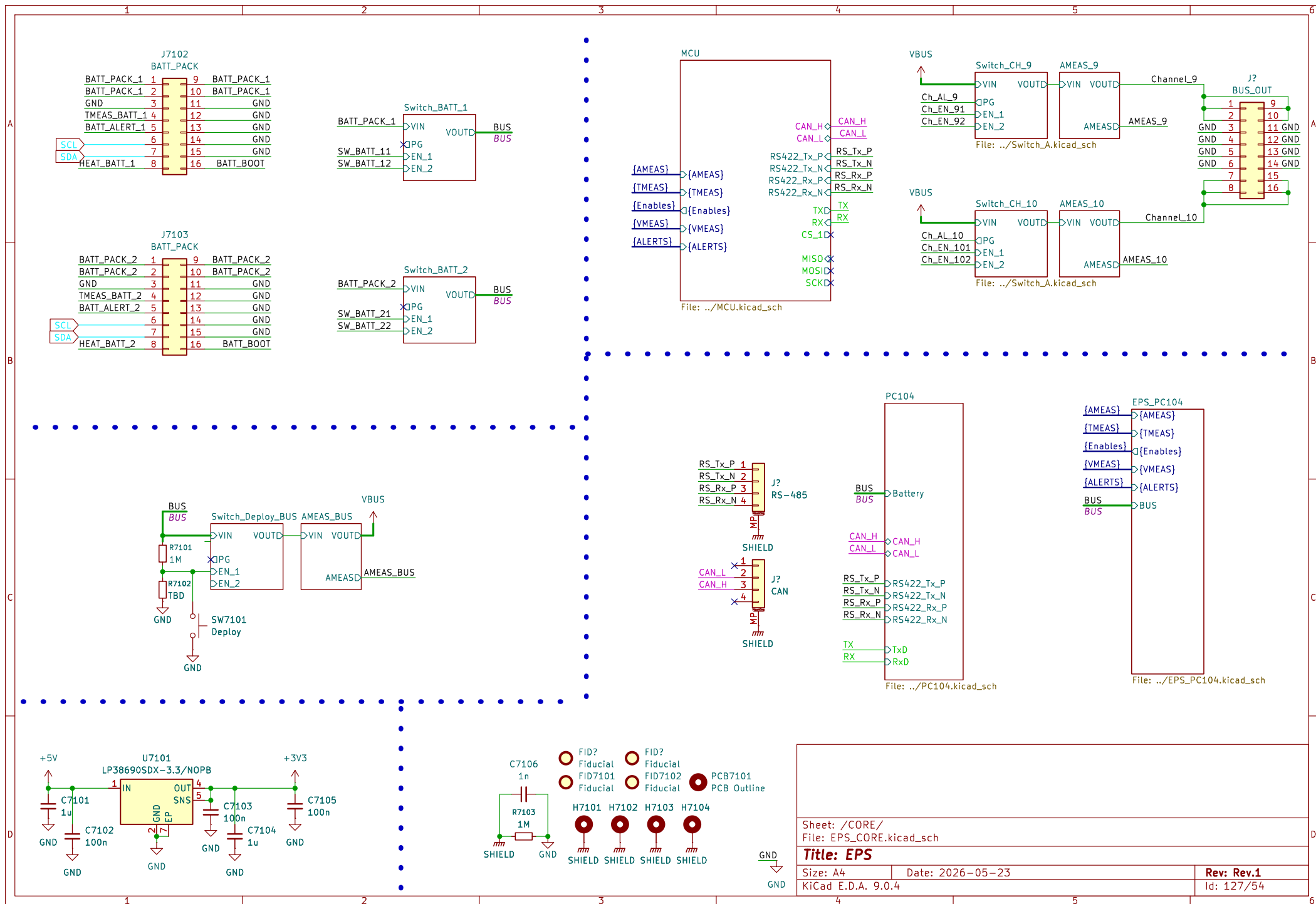
Size: A4 Date: 2026-05-23

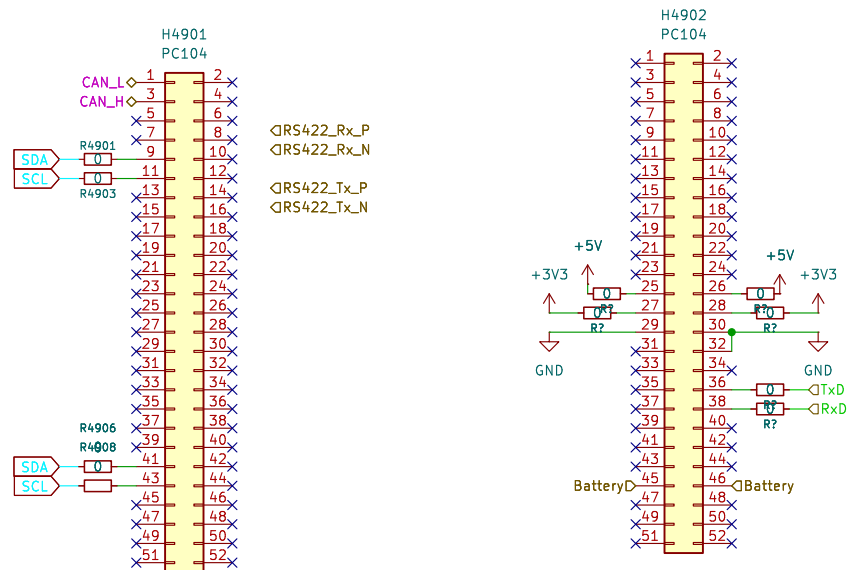
KiCad E.D.A. 9.0.4

Rev: Rev.1

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Sheet: /CORE/PC104/
File: PC104.kicad_sch

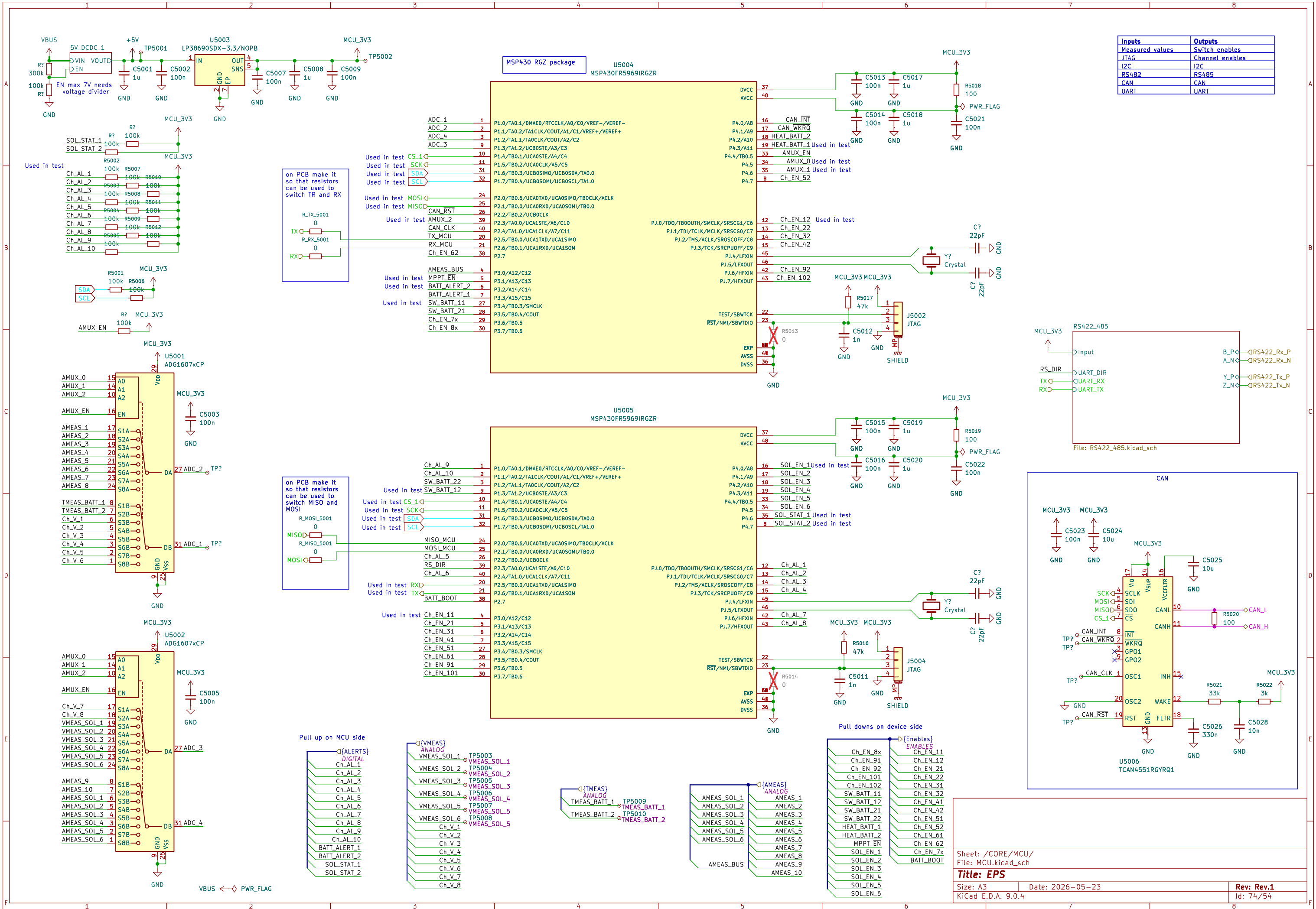
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Size: A4 Date: 2026-05-23

KiCad E.D.A. 9.0.4

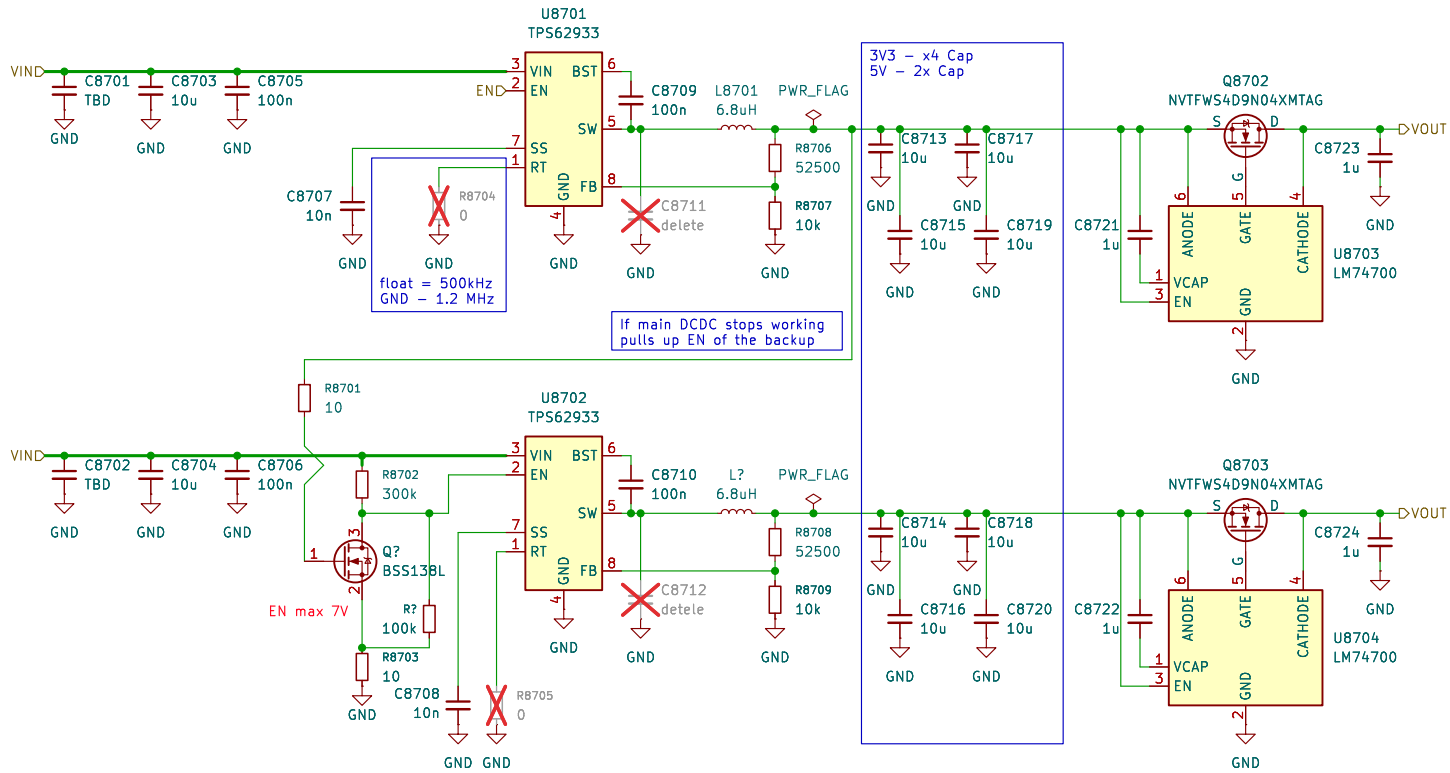
Rev: Rev.1

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Inputs	Outputs
BUS Voltage	Set voltage

Output voltage
5V
 $10000 \cdot (5V - 0.8) / 0.8 \Rightarrow R_{xx03}, R_{xx00} = 52500$



Sheet: /CORE/MCU/5V_DCDC_1/
File: DCDC_ADJUCTABLE.kicad_sch

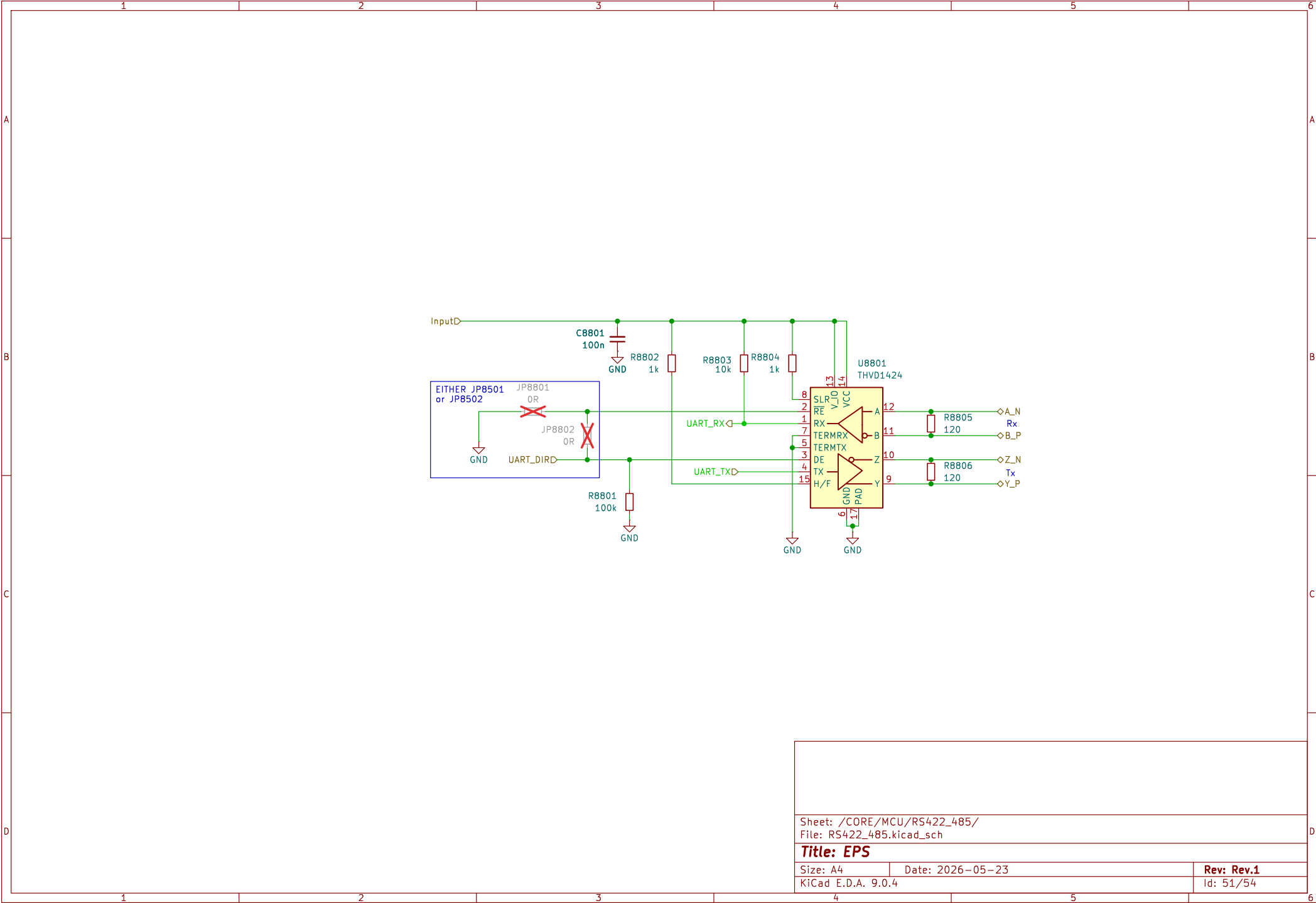
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KiCad E.D.A. 9.0.4

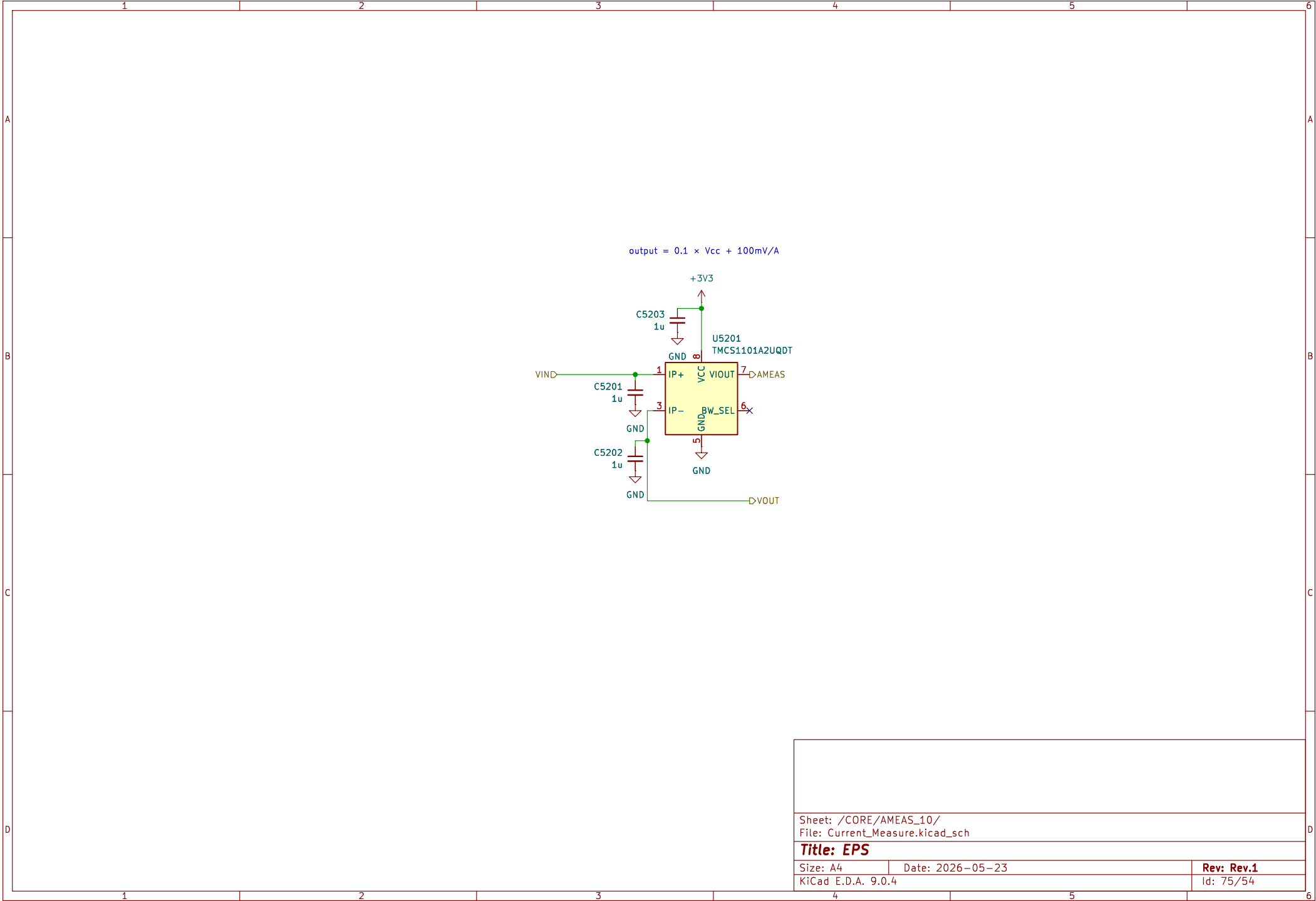
Rev: Rev.1

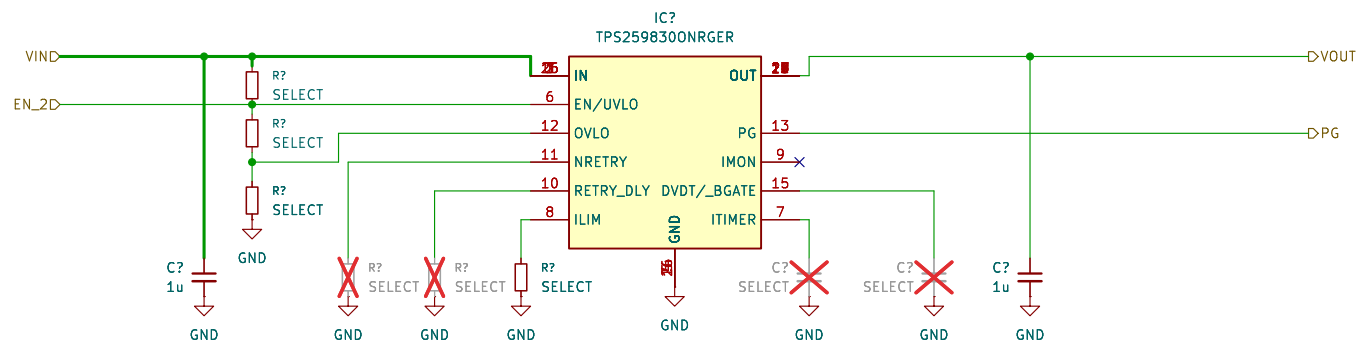
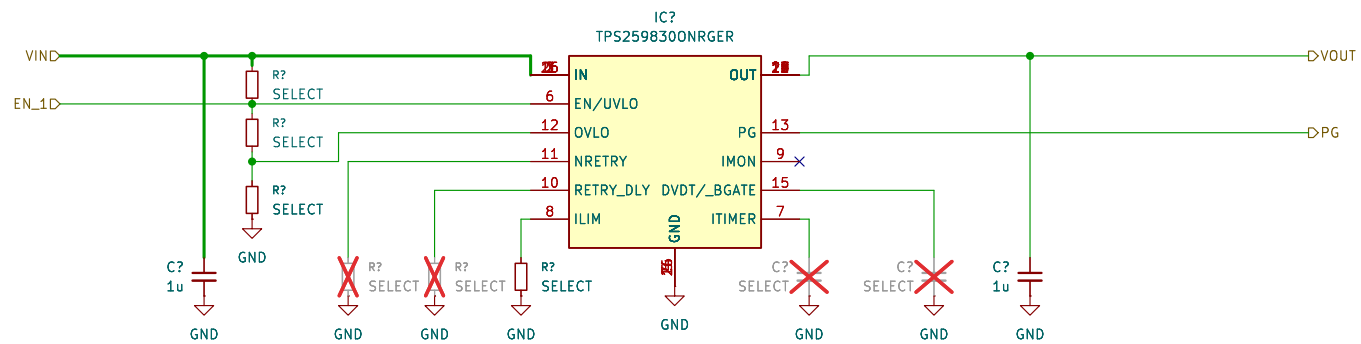
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Title: *EPS*





Sheet: /CORE/Switch_CH_9/
File: Switch_A.kicad_sch

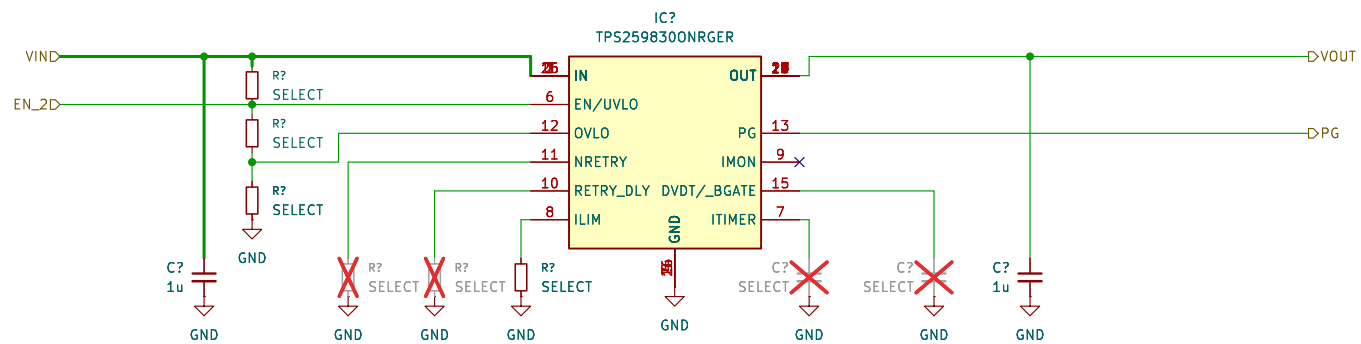
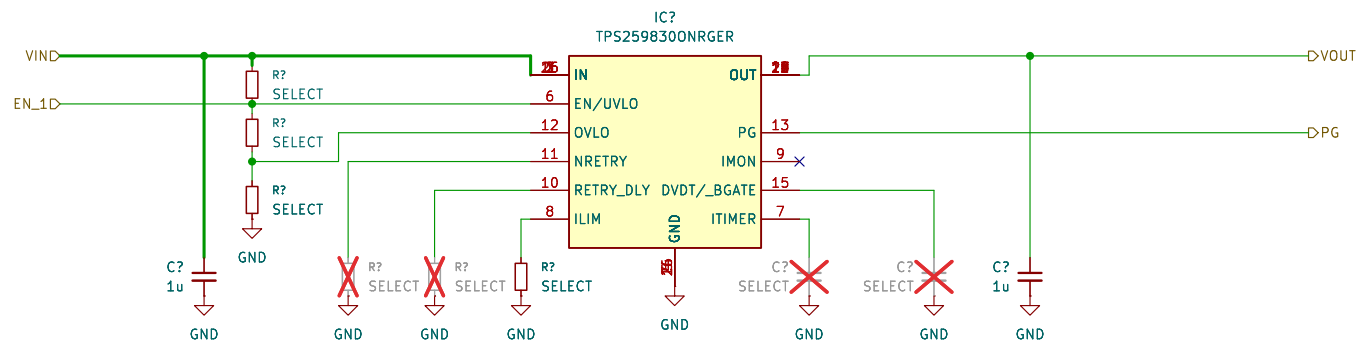
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Size: A4	Date: 2026-05-23
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Sheet: /CORE/Switch_CH_10/
File: Switch_A.kicad_sch

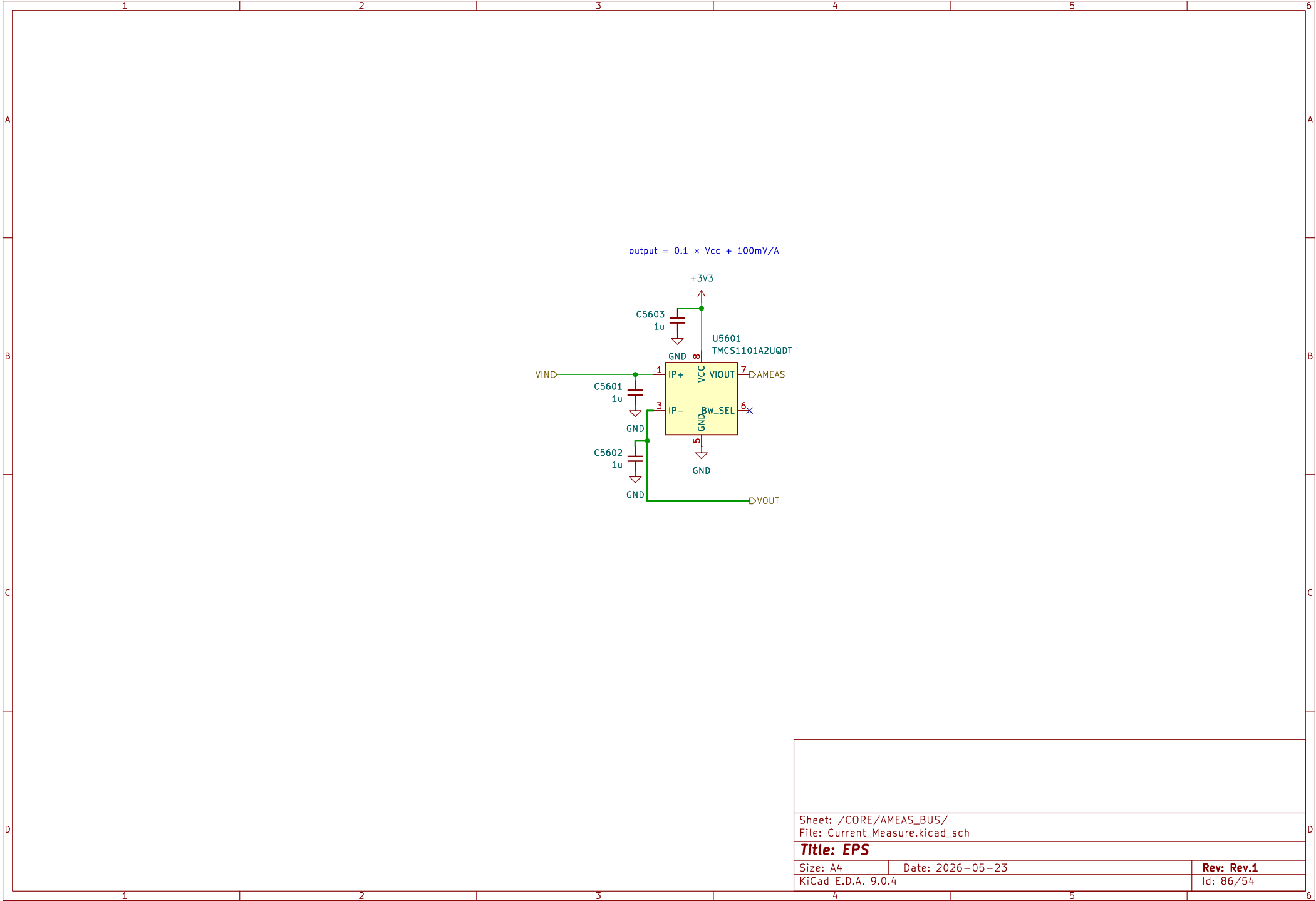
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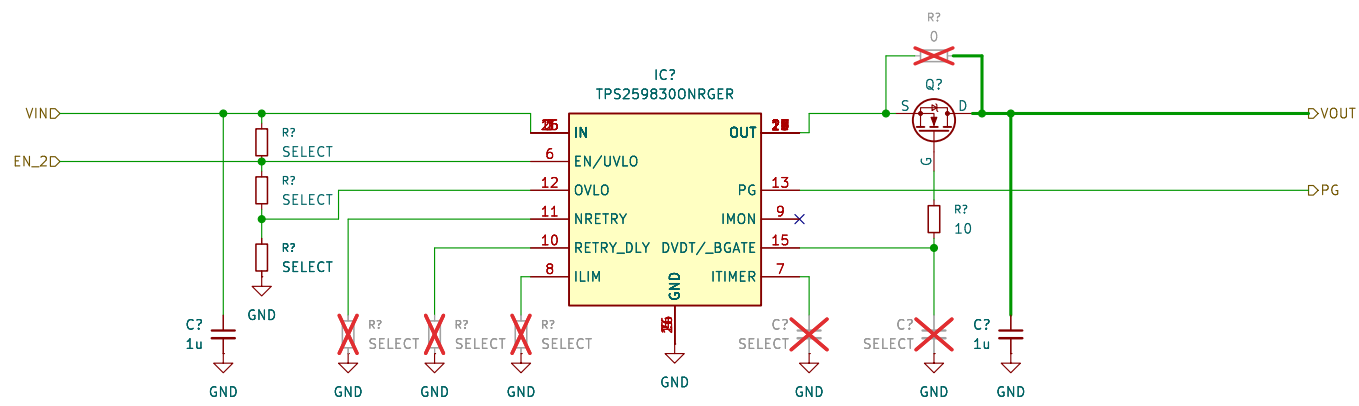
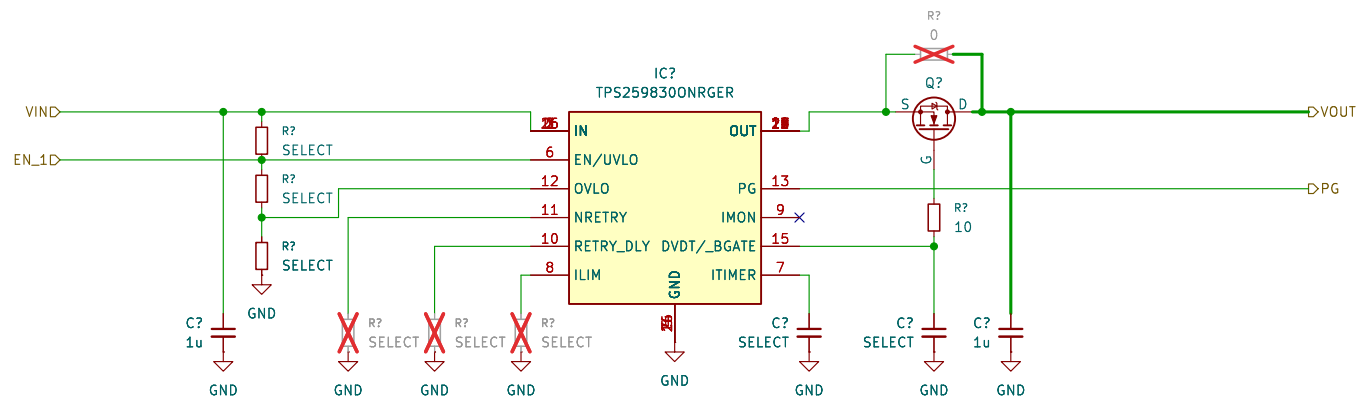
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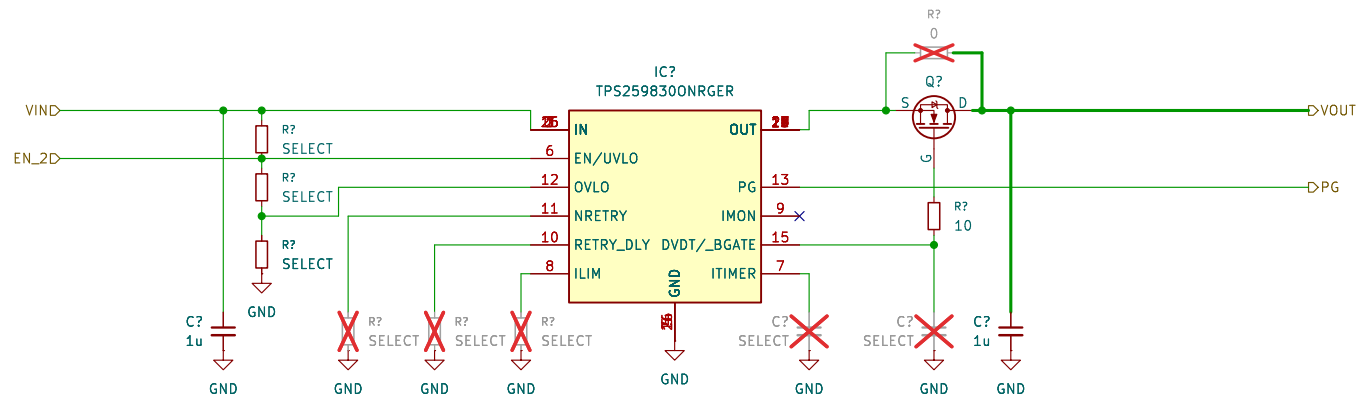
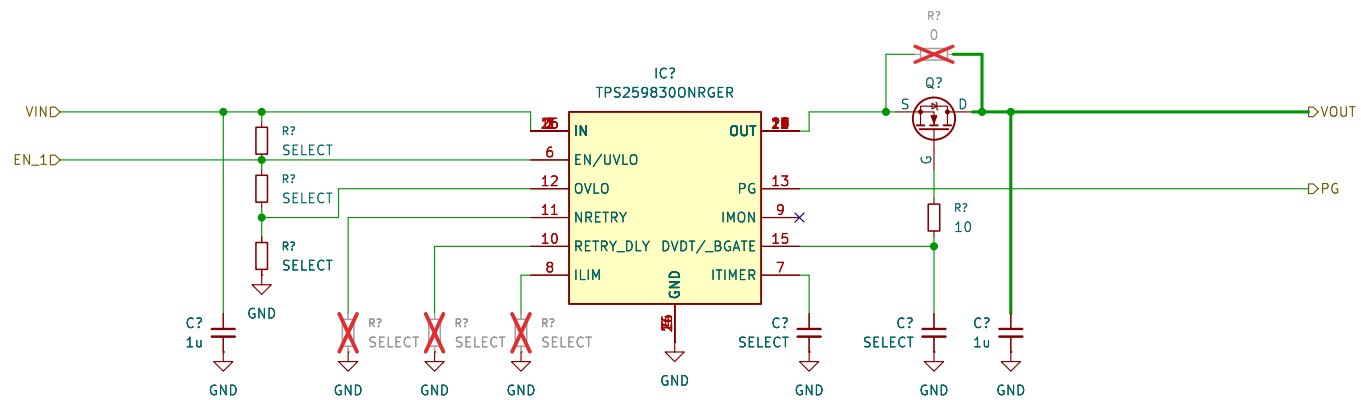
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Rev: Rev.1

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Sheet: /CORE/Switch_BATT_2/
File: Switch_B.kicad_sch

Title: *EPS*

Size: A4

Date: 2026-05-23

KiCad E.D.A. 9.0.4

Rev: Rev.1

Id: 111/54

